

# IS THERE A PLATONIC TREE?

## CALIBRATING LATENT HYPERBOLICITY

### IN FOUNDATION MODELS

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Paper under double-blind review

#### ABSTRACT

Hyperbolic methods for representation learning rest on a premise we call latent hyperbolicity: standard models are already tree-like, because their class geometry scores a low Gromov  $\delta$ . We show that a low raw reading is what high dimension, an anisotropic spectrum and a maximum over sampled quadruples produce on their own, and we build an instrument that reads every score as an excess over a random cloud of the same shape, ranks it against 200 matched replicates, and adds a depth test whose false-alarm rate and power are measured on real clouds. Across 12 vision backbones, 6 datasets and 16 text models, the premise does not survive: on image features the excess sits within null noise in most cells (model-dataset pairs). Class centroids do carry clustered structure in 49 of 72 cells, but a star already produces it; hierarchy above the superclasses is certified in only 4 of 12 ImageNet backbones, by a test that never fires on randomized hubs, and a backbone trained in hyperbolic space shows the same clustering, no detected depth, and embeddings that never leave the near-flat regime. The trees are moderately shared: the naive comparison manufactures an island, and once the clustering cut is controlled every recipe recovers the human taxonomy partially, with the self-supervised tree living in angles rather than distances. Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical and moderately shared; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry.

## 1 INTRODUCTION

Semantic knowledge is hierarchical, and hyperbolic space is the natural geometric home for hierarchies (Nickel & Kiela, 2017; Sarkar, 2011; Ganea et al., 2018; Chami et al., 2019). A line of work therefore imposes it on representations at training time, from vision to vision-language and language (Khrulkov et al., 2020; Ermolov et al., 2022; Atigh et al., 2022; Bdeir et al., 2024; Desai et al., 2023; Pal et al., 2025; Chen et al., 2022; He et al., 2025; Sinha et al., 2024). These methods cite a premise, which we call *latent hyperbolicity*: standard models already organize concepts in a tree-like metric structure. The evidence is a single number, the Gromov  $\delta$  of their features:  $\delta$  is zero for a metric tree and grows as a metric departs from one, and the measured values are low, on CNN and ViT features (Khrulkov et al., 2020; Bdeir et al., 2024), on token embeddings (Yang et al., 2025) and on word vectors (Tifrea et al., 2019). Those readings are taken at the sample level, on individual images or tokens rather than on classes, and they are never calibrated: a value is called low on its own, without asking what a structureless cloud of the same dimension and spectrum would score. Is the premise right, and what does it license? Figure 1 shows why a low reading is not enough: a structureless cloud, a star of clusters without depth and a tree with depth cast the same shadow.

figures/fig1\_concept.pdf (conceptual figure, drawn by the authors; same-size placeholder)

Figure 1: **Plato’s cave: a random cloud, a star and a tree cast the same shadow.** An observer who sees only shadows cannot tell which object cast them, and a raw  $\delta$  is such a shadow: points with no structure, clusters around a hub, and a nested hierarchy all read as low  $\delta$ . The instrument reads the objects instead of the shadow: no structure beyond the null in the cloud, clusters without depth in the star, clusters with depth in the tree.

Why is the shadow low? Gromov  $\delta$  assigns zero to trees (Gromov, 1987), but three artifacts push it down without any hierarchy. High-dimensional clouds concentrate toward equidistance (Beyer et al., 1999; Aggarwal et al., 2001), which is a degenerate star tree: we call this the *dimension confound*, the geometric face of the width confounder of Gröger et al. (2026). Anisotropic spectra mimic low-dimensional behavior: the *spectrum confound*. And the supremum over sampled quadruples is a one-quadruple extreme that does not converge (Fournier et al., 2015): the *statistic confound*. The three are visible in real models: GPT-2 M sits at its matched null despite a low raw reading, and classic sentence embedders share their raw reading with vision-contrastive models while carrying none of their structure.

We build the instrument that reads  $\delta$  correctly. It calibrates the estimator on reference geometries, expresses every reading as an *excess* over a null cloud matched in dimension and covariance spectrum, ranks that excess against matched replicates, and adds a *depth test* whose false-alarm rate and power are measured on real clouds. With it we ask of geometry what convergence work asks of similarity (Huh et al., 2024; Gröger et al., 2026): what survives calibration, and is it shared? They calibrate similarity across models; we calibrate geometry within a model, and comparing trees across models needs a calibration of its own.

**The answer has three parts.** On image features the excess sits within null noise in most cells and is a fraction of the null where it survives, so the premise does not survive calibration as stated. On class centroids, clustered structure is genuine in 49 of 72 cells, but a star already produces it. Hierarchy above the superclasses is certified in 4 of 12 ImageNet backbones by a test that never fires on real clouds with randomized hubs, and a backbone trained in hyperbolic space shows the same clustering and no detected depth. The trees are moderately shared: the naive comparison manufactures an island, and once the clustering cut is controlled a moderate gap remains, every recipe recovers the human taxonomy partially, more so with supervision, and the self-supervised structure lives in angles.

#### Our contributions are:

1. **The instrument.** Estimator calibration, spectrum-matched nulls read as corrected ranks, and a depth test with measured false-alarm rate and power.
2. **The census.** Clustered structure in most models, hierarchy certified in a few, and a hyperbolic backbone as the control for imposing the geometry.
3. **The map of trees.** The island is an artifact of the cut; sharing is graded and supervision-dependent, and the self-supervised tree is angular.
4. **Consequences for practice.** What to measure before imposing curvature, and which zero-cost readout collects what is there.

## 2 RELATED WORK

Reading  $\delta$  against a matched reference has precedent in network science, where the  $\delta$  of real graphs is compared with that of random graphs of comparable size (Narayan & Saniee, 2011; Kennedy et al., 2013; Adcock et al., 2013); we bring the idea to embedding clouds, where the reference must match dimension and spectrum. The analyses of learned representations characterize Euclidean structure and do not test for tree-likeness: representation similarity, intrinsic dimension and neural collapse (Kornblith et al., 2019; Ansuini et al., 2019; Pope et al., 2021; Pappan et al., 2020). Closest to us is the work on representational convergence: the Platonic Representation Hypothesis (Huh et al., 2024) argues that models converge toward shared representations, and its cross-modal convergence has been stress-tested at scale (Koepke et al., 2026); Gröger et al. (2026) show that permutation-calibrated similarity removes most of that convergence while local neighborhood structure survives, and Park et al. (2024) read concept hierarchy in language models from the orthogonality of concept directions. We are the geometric complement of this program: the same calibration applied within a model instead of across models, with a depth test measured on real clouds, a hyperbolic backbone as the control for imposing the geometry, and a calibration of the tree-to-tree comparison itself.

## 3 METHODOLOGY

This section fixes what is measured, how it is estimated and normalized, what it is compared with, how depth above the clusters is tested, and how trees are compared across models.

### 3.1 BACKGROUND: GROMOV $\delta$

Tree-likeness is measured on quadruples. Take any four points  $w, x, y, z$  of a metric space with distance  $d$ . They can be paired in three ways, and each pairing has a sum of two distances:

$$d(w, x) + d(y, z), \quad d(w, y) + d(x, z), \quad d(w, z) + d(x, y). \quad (1)$$

Sort the three sums as  $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$ . In a tree the two largest sums are always equal, because the two pairings that cross the trunk traverse the same path, so the four-point defect  $(S_{\max} - S_{\text{mid}})/2$  is zero; the further a metric is from a tree, the larger the defect can become. Gromov  $\delta$  is the worst case, the supremum of the defect over all quadruples (Gromov, 1987). A metric tree has  $\delta = 0$ . Euclidean space has no bound at all: enlarging a square enlarges its defect in proportion. Hyperbolic space sits in between, with a bounded defect whose value for the hyperbolic plane is known,  $\ln 2$  under this definition (Nica & Špakula, 2016).

**Definition 1** (four-point defect). For a quadruple  $(w, x, y, z)$  with the pairing sums of Eq. 1 sorted as  $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$ , the four-point defect is  $\delta(w, x, y, z) = \frac{1}{2} (S_{\max} - S_{\text{mid}})$ , and the Gromov  $\delta$  of a metric space is the supremum of the defect over its quadruples.

### 3.2 ESTIMATION AND NORMALIZATION

Three steps turn  $\delta$  into a number that can be compared across models. First, a thousand centroids have far more quadruples than can be enumerated, so the defect is computed on half a million random quadruples. Second, the largest defect in that sample is a poor estimate of the supremum: it belongs to a single extreme quadruple and keeps growing as more quadruples are drawn (Fournier et al., 2015). We therefore read the 99.9th percentile of the sampled defects,  $\hat{\delta}_{99.9}$ , the value that 99.9 per cent of them fall below: it keeps the tail of the defects, the intent of the supremum, but it is set by hundreds of quadruples rather than by one. It also does not move with the sample size: across quadruple budgets the ImageNet reading drifts by less than half its own spread (Table 2). Third, the defect grows with the size of the point set, so we divide by the diameter, the largest pairwise distance:

$$\delta_{\text{norm}} = \frac{\hat{\delta}_{99.9}}{\text{diam}}. \quad (2)$$

On this scale a tree reads zero, and a hyperbolic region of radius four or more reads below a sphere (Table 3). What a structureless cloud reads on it is the next question.

**Definition 2** (reading of record). The reading of record of a cloud is  $\delta_{\text{norm}}$  of Eq. 2, with  $\hat{\delta}_{99.9}$  computed over half a million sampled quadruples per seed and averaged over ten seeds.

**Lemma 1** (range bound). *Let  $d_{\min}$  and  $d_{\max}$  be the smallest and the largest pairwise distance of a finite metric space. For any quadruple the four-point defect is at most  $d_{\max} - d_{\min}$ , hence  $\delta_{\text{norm}} \leq 1 - d_{\min}/d_{\max}$ .*

**Corollary 1** (dimension confound). *For an iid Gaussian cloud of  $n$  points in dimension  $d$ ,  $d_{\min}/d_{\max} \rightarrow 1$  as  $d \rightarrow \infty$  (Beyer et al., 1999; Aggarwal et al., 2001), so  $\delta_{\text{norm}} \rightarrow 0$  without any structure.*

*Remark.* An anisotropic spectrum changes the effective dimension of the cloud and moves the same bound: the spectrum confound. A supremum over sampled quadruples is a maximum that grows with the sample: the statistic confound.

The proofs are in Appendix A. The calibration on reference geometries confirms the bound in practice: the estimator recovers the tree zero and the  $\ln 2$  constant of the hyperbolic plane, spheres stay high, and iid Gaussians drift down with  $d$  (Table 3).

### 3.3 SPECTRUM-MATCHED NULL, EXCESS AND GENUINENESS

**Definition 3** (Haar null). Let  $X = U\Sigma V^\top$  be the singular value decomposition of the centered cloud. A null replicate is  $X' = Q\Sigma V^\top$  with  $Q$  Haar-distributed on the orthogonal group, so  $X'$  has the singular values of  $X$  and structureless coefficients.

In plain words, the null is a cloud with the same shape as the real one and no structure inside it. Gaussian coefficients reproduce the spectrum only in expectation, fall short of it when  $n$  is small relative to  $d$ , and inflate the excess of low-rank clouds. The Haar construction reproduces the sample spectrum exactly, so it is the record, and the Gaussian reading is reported alongside it in Table 4.

#### Excess, rank and genuineness.

**Definition 4** (excess). With  $K$  null replicates  $X'_1, \dots, X'_K$  of Definition 3, the excess of a cell is

$$e(X) = \delta_{\text{norm}}(X) - \frac{1}{K} \sum_{k=1}^K \delta_{\text{norm}}(X'_k). \quad (3)$$

The primary evidence is the rank of the reading among its replicates and the left-tail  $p$  that the rank gives,

$$r = \#\{k : \delta_{\text{norm}}(X'_k) > \delta_{\text{norm}}(X)\}, \quad p = \frac{1 + K - r}{K + 1}, \quad K = 200. \quad (4)$$

**Definition 5** (genuine). Order the  $m$  left-tail  $p$ -values of a census as  $p_{(1)} \leq \dots \leq p_{(m)}$ . A cell is genuine when its  $p$  is at or below the Benjamini–Hochberg threshold at level five per cent,

$$p \leq p_{(i^*)}, \quad i^* = \max\{i : p_{(i)} \leq 0.05 i/m\}, \quad (5)$$

and the word is used in this sense only.

A negative genuine excess certifies clustered structure. The word tree-like is reserved for the depth test of Section 3.4. A positive excess means less tree-like than chance given the spectrum. Figure 2 shows the two readings together: in panel (a) the null of each ImageNet backbone moves with its dimension inside a band of two null spreads, so the gap between the filled and the hollow marker, not the level of the filled one, is the reading; in panel (b) two cells with the same raw reading receive opposite verdicts, because one null sits at the reading and the other far above it.

### 3.4 DEPTH TEST

**Definition 6** (hub null and excess B). Given a frame that assigns each centroid  $x_i$  to a cluster  $k(i)$  with hub  $h_{k(i)}$ , the mean of its cluster, write  $x_i = h_{k(i)} + o_i$ . The hub null keeps every offset  $o_i$  and replaces the hub configuration by a Haar resample with the hubs' spectrum,  $x'_i = h'_{k(i)} + o_i$ ; the excess of Eq. 3 read against this null is  $e_B$ , and it measures hierarchy above the clusters of the frame only.

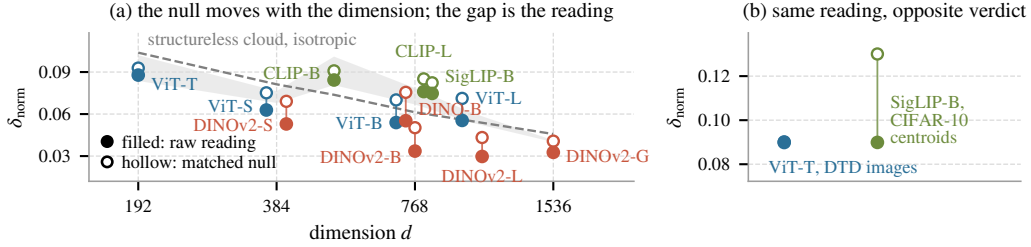


Figure 2: **The raw reading cannot be read on its own.** (a) Each ImageNet backbone’s raw reading (filled, labeled) and what a structureless cloud of the same dimension and spectrum reads (hollow), joined by the excess; the shaded band is the null mean plus or minus two null spreads across dimension, and the dashed line the isotropic Gaussian reference. (b) Two cells with the same raw reading and opposite verdicts: on ViT-T image features the null sits at the reading, on SigLIP-B centroids far above it.

**Definition 7** (matched star). The matched star of a cloud keeps the real clusters, each as a Haar sample with the cluster’s own covariance, and places their hubs as Gaussian draws at the real hub radius, so that it carries the clustering of the real cloud and no hierarchy above it. Its spectrum-matched-hub variant draws the hubs as a Haar resample of the real hubs instead.

With  $e_B$  read on the real cloud and on its matched star, the depth statistic and its standardized form are

$$\text{depth} = e_B(X) - e_B(X_{\text{star}}), \quad z = \text{depth} / \text{s.d.}, \quad (6)$$

where the spread is taken over star seeds and null replicates, and a cloud is certified as more hierarchical than a star when  $z \leq -2$ . The test is needed because of the star caveat: a pure star of clusters with no hierarchy already passes the census of Definition 5, so the excess measures how classes are arranged and not whether the arrangement has depth. The frame is the WordNet cut on ImageNet and the coarse labels on CIFAR-100. The false-alarm rate and the power of the test are measured on real clouds in Section 5.3.

### 3.5 COMPARING TREES ACROSS MODELS

Each model’s hierarchy is a dendrogram over the shared class centroids, and two hierarchies are compared by the adjusted Rand index between their cut partitions, by the correlation between their cophenetic distances, and by their agreement on which pair of a random class triplet merges first, a measure that needs no cut. A configuration of metric and linkage is selected by a criterion fixed without reference to the agreements being compared: the highest cophenetic fidelity among configurations whose reference cuts are non-degenerate for every model, applied per dataset because degeneracy is a property of the dataset. Local agreement is read with the permutation calibration of Gröger et al. (2026): permutations of the class correspondence give the null distribution of mutual-kNN agreement and of centroid CKA, the upper quantile of that distribution gives the threshold, and the calibrated score is the excess over the threshold rescaled to keep the maximum at one.

### 3.6 PROJECTION AND TASKS

A parameter-free map centers the features, scales the 95th-percentile norm to radius  $1/\sqrt{2}$  and preserves directions, so that Euclidean, cosine and Poincaré readouts can be compared on the intact representation. Nearest-centroid classification and 5-way 5-shot episodes score the three readouts on every cell of the task grid. The next section fixes the models, the data and the protocol on which the instrument is run.

## 4 EXPERIMENTAL SETUP

**Models.** Twelve vision backbones form the census: four supervised ViTs trained on ImageNet-21k, ViT-T to ViT-L, DINO-B, four DINOv2 sizes from S to G, CLIP-B, CLIP-L and SigLIP-B. Two

self-supervised ResNet-50 backbones read the same census in the appendix. Two ViT-B trained on ImageNet-1k leaf labels, DeiT-B and the augreg ViT-B, serve as the label-set control. MERU and its Euclidean CLIP twin, in three sizes, serve as the hyperbolic-backbone control. Sixteen text models read the thousand ImageNet class names: four GPT-2 sizes, three Pythia sizes, two OLMo sizes and seven sentence embedders. Parameter counts and the full panel are in Table 5.

**Datasets and cells.** The vision census reads six class sets: ImageNet and CIFAR-100, which carry a real hierarchy, CIFAR-10 and DTD, and the flat FMNIST and MNIST. Text is read on the ImageNet class names, on DBpedia Classes, a three-level ontology independent of WordNet, and on HierarCaps, a four-level caption hierarchy. A cell is one pair of model and class set. The premise is read at the sample level, on per-image features, roughly a thousand stratified training images per cell for CIFAR-100 and DTD; hierarchy is read at the class level, on class centroids.

**Extraction.** Vision backbones give the encoder’s default pooled or CLS embedding, and class centroids average a hundred cached training images per class on ImageNet and the full cached split on the other datasets. Causal language models give the last-token hidden state of “a photo of a {class}”, extracted one prompt at a time, because left-padded batching alters GPT-2’s hidden states; embedders use their recommended pooling, and the CLIP and SigLIP text towers their text encoders (Appendix B.11).

**Replicates, seeds, budgets and frames.** Every cell is read against 200 Haar replicates, with half a million sampled quadruples per seed and ten seeds for the real cloud, and genuineness is decided by Benjamini–Hochberg over the 72 vision cells and, separately, over the text census. The depth test uses the WordNet cut into thirty superclasses on ImageNet and the twenty coarse labels on CIFAR-100, a pre-specified balanced frame as the control, a sweep over the number of clusters, and ten star seeds under the hub null. Its false-alarm rate and power are measured on synthetic hierarchies at ImageNet’s class count and on the real ImageNet clouds with implanted two-level trees. With the setup fixed, we read the premise where it is read.

## 5 RESULTS

### 5.1 THE PREMISE AT THE SAMPLE LEVEL

**The premise does not survive calibration.** On per-image features the raw reading lies in the low band the literature reports. Calibrated against the matched null, it is not genuine in 14 of 24 cells. Where it is genuine, the excess removes at most 39 per cent of the null reading, and the largest figure belongs to DINOv2-G on CIFAR-100 images. The genuine cells belong mostly to the DINOv2 family, whose structure is the angular tree of Section 5.4, which cosine collects. Table 6 gives the reading per cell, with the raw supremum the literature reports beside it.

### 5.2 CLUSTERED STRUCTURE AT THE CLASS LEVEL

**Almost every cell is below its null.** On class centroids the excess is negative in 70 of 72 cells. It is genuine in 49 of 72. Table 1 gives the census of record row by row, and Figure 3 shows which markers fall outside the null band.

**The count survives resampling.** On ImageNet and CIFAR-100, the two sets with a real hierarchy, the count is 18 of 24. Folding image resampling and estimator noise into the null, or repeating the census on thirty resampled centroid sets, leaves between 42 and 47 of the 72 cells genuine, and the cells that move are on the transfer sets (Table 2).

**The structure belongs to no single family.** Every family contributes genuine cells. The smallest backbone, ViT-T, is genuine on DTD only. The flat datasets sit near the resolution limit of ten classes, and a class-count control on ImageNet subsets shows that the reading follows the hierarchy of the labels rather than their number, as Table 2 reports. Two self-supervised ResNet-50 backbones read the same clustered structure as the ViTs (Table 4).

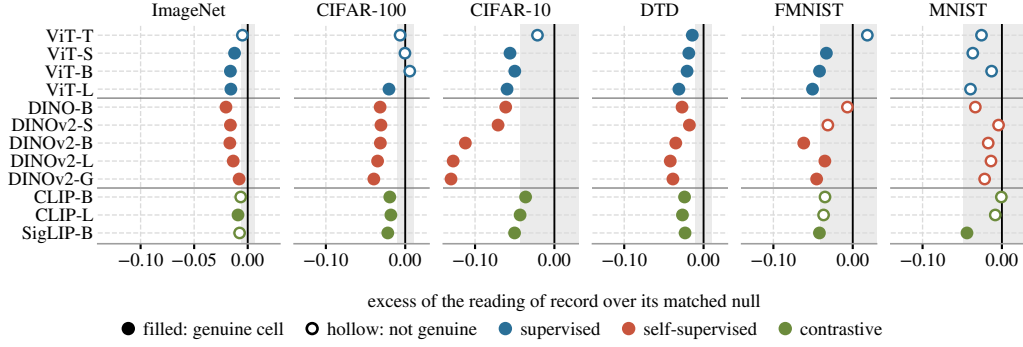


Figure 3: **Clustered structure is the rule at the class level.** Excess of the reading of record over its matched null, one panel per dataset, one marker per backbone; filled markers are genuine cells, and the shaded band around zero spans two median null spreads. Almost every marker sits left of zero, most are filled, the flat datasets scatter around zero, and the DINOv2 rows reach furthest left on CIFAR-10.

| model    | family | $\hat{\delta}_{99.9}$ IN | excess IN      | $r/200$ ( $p$ ) | exc. C100      | exc. C10       | $\rho_{WN}$ |
|----------|--------|--------------------------|----------------|-----------------|----------------|----------------|-------------|
| ViT-T    | sup.   | 0.088                    | $-0.005^\circ$ | 169 (.159)      | $-0.006^\circ$ | $-0.021^\circ$ | +0.52       |
| ViT-S    | sup.   | 0.063                    | $-0.012$       | 200 (.005)      | $-0.000^\circ$ | $-0.056$       | +0.50       |
| ViT-B    | sup.   | 0.054                    | $-0.016$       | 200 (.005)      | $+0.006^\circ$ | $-0.050$       | +0.49       |
| ViT-L    | sup.   | 0.055                    | $-0.016$       | 200 (.005)      | $-0.020$       | $-0.060$       | +0.53       |
| DINO-B   | SSL    | 0.055                    | $-0.020$       | 200 (.005)      | $-0.032$       | $-0.061$       | +0.36       |
| DINOv2-S | SSL    | 0.053                    | $-0.016$       | 200 (.005)      | $-0.031$       | $-0.071$       | +0.22       |
| DINOv2-B | SSL    | 0.034                    | $-0.017$       | 200 (.005)      | $-0.031$       | $-0.112$       | +0.18       |
| DINOv2-L | SSL    | 0.030                    | $-0.014$       | 200 (.005)      | $-0.035$       | $-0.128$       | +0.18       |
| DINOv2-G | SSL    | 0.033                    | $-0.008$       | 200 (.005)      | $-0.040$       | $-0.130$       | +0.20       |
| CLIP-B   | contr. | 0.084                    | $-0.007^\circ$ | 172 (.144)      | $-0.019$       | $-0.036$       | +0.57       |
| CLIP-L   | contr. | 0.076                    | $-0.009$       | 197 (.020)      | $-0.018$       | $-0.043$       | +0.59       |
| SigLIP-B | contr. | 0.075                    | $-0.008^\circ$ | 189 (.060)      | $-0.022$       | $-0.050$       | +0.58       |

Table 1: **Clustered structure is the rule at the class level.** One row per backbone.  $\hat{\delta}_{99.9}$ : raw ImageNet reading; excess:  $\hat{\delta}_{99.9}$  minus the mean of 200 Haar spectrum-matched null replicates;  $r/200$ : replicates above the real value, with its left-tail  $p$ ; genuine: Benjamini–Hochberg-corrected  $p \leq 0.05$  over the 72 cells;  $^\circ$ : not genuine; bold: less clustered than the null.  $\rho_{WN}$ : Spearman correlation of inter-centroid and WordNet distances. The same reading on per-image features is genuine in 10 of 24 cells (Table 6). Under image resampling and estimator noise the genuine count is 42–47 of 72 (Table 2). Other datasets and the four null  $\times$  statistic verdicts: Table 4; task columns: Table 14.

**The excess is small in absolute terms and large against the null.** On ImageNet centroids the excess lies between 1 and 36 null spreads below the null mean. The absolute values are in Table 1.

**The star caveat.** A synthetic star of thirty clusters with no hierarchy scores an excess as deep as any real cell, reaching  $-0.115$  under the Haar null. A negative excess therefore says clustered, not deep, and the depth of Section 5.3 needs the test of Definition 7 (Table 3).

**Neural collapse is the limit, not the reading.** Neural collapse predicts that the class means of a supervised classifier converge to a simplex equiangular tight frame (Pappayan et al., 2020), and an exact frame is the extreme case of the star caveat. The arrangement we certify is not that frame: class means cluster around superclass hubs that a frame would erase, cutting the centroid dendrograms recovers the CIFAR-100 superclasses, and the same clustered structure appears without class labels and in causal language models, where neural collapse is not stated (Table 7). Scale deepens the

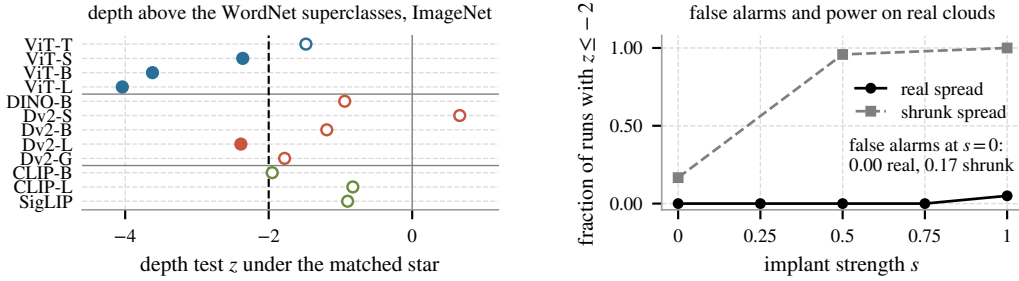


Figure 4: **The depth test certifies hierarchy above the superclasses in 4 of 12 ImageNet backbones, never fires on randomized hubs, and misses implanted depth at ImageNet’s noise level.** (a) Depth  $z$  per backbone under the matched star; filled markers are certified ( $z \leq -2$ ). (b) Detection rate against implant strength on the real clouds: zero at  $s = 0$  and low at  $s = 1$  with the real within-cluster spread, full once that spread is shrunk into the validated range (dashed).

excess for one family only: the DINOv2 rows of Figure 3 reach further left with size on CIFAR-10 and, weakly, on CIFAR-100. No other family shows a trend with scale, and training moves the raw reading within an architecture: fine-tuning on a task without class hierarchy raises it, and it falls across transformer depth, as Table 7 reports.

### 5.3 DEPTH ABOVE THE LABELLED CLUSTERS

**Depth is certified in four of twelve.** On ImageNet the real cloud is more hierarchical than its matched star in 4 of 12 backbones, ViT-S, ViT-B, ViT-L and DINOv2-L, and never less. Under the spectrum-matched-hub star the certified set is the same four, with  $z$  from  $-2.0$  to  $-3.7$ . Figure 4a shows the four filled markers, and Table 8 gives both stars per backbone.

**The test raises no false alarm on real clouds.** With the hub arrangement of every real cloud randomized, the test fires in 0 of 60 runs. For the four certified backbones the same clusters with random hubs never fire, 0 of 20, so their verdict comes from the real hub arrangement.

**The test is weak at ImageNet’s noise level.** A clean implanted two-level tree is detected in almost none of the runs, a power of 0.05. The reason is the noise level: the within-cluster spread of the real clouds is 1.3 to 3.9 times their between-hub spread, outside the range the synthetic sweep covered. Figure 4b shows the detection rate staying at zero below full strength with the real spread, and Table 9 gives the power by strength. The eight other backbones are not detected, which is not the same as not hierarchical. The certified set shifts with the choice of frame. CIFAR-100, with a hundred centroids in twenty clusters, lies outside the validated regime, and its readings are reported without certification. An earlier isotropic star produced false alarms and is withdrawn. A variant with the within-cluster offsets shrunk into the validated range detects all twelve backbones, but it also fires at zero strength for two contrastive backbones, so it is a diagnostic of power and not a substitute test.

**Leaf-label supervision can produce the depth.** The augreg ViT-B trained on ImageNet-1k leaf labels is certified, at  $z = -3.03$  under the Gaussian-hub star and under the spectrum-matched one alike. It is also aligned with WordNet, at a Spearman correlation of  $+0.52$ , like the ImageNet-21k ViTs.

**But it does not guarantee it.** DeiT-B, trained on the same leaf labels with another recipe, is not certified and is nearly unaligned, at  $+0.08$ . It nevertheless recovers the CIFAR-100 superclasses from its dendrograms at an adjusted Rand index of 0.56. Hierarchical label sets are therefore not the explanation of the certified ViTs’ depth, and alignment with the human taxonomy is recipe-dependent even among leaf-supervised ViTs (Table 8).

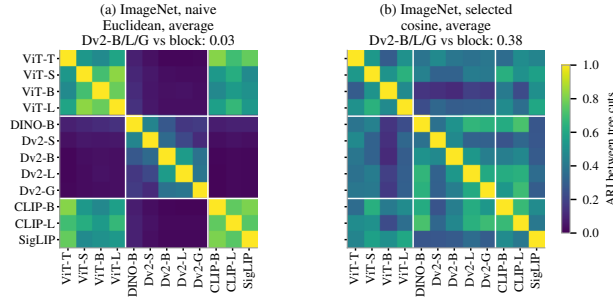


Figure 5: **The categorical island is a chaining artifact; about a third of the within-block agreement is missing under most configurations.** Pairwise ARI between the dendrogram cuts of the 12 vision models on ImageNet under the naive configuration (a) and the selected one (b), with the mean DINOv2-vs-block ARI above each panel.

**Imposing the geometry does not create detected depth.** MERU embeds images and text on the Lorentz hyperboloid with an entailment objective, and its released Euclidean twin shares backbone, data and recipe minus the hyperbolic lift (Desai et al., 2023). The two read the same excess on ImageNet images, MERU’s reading in its own Lorentz metric equals its Euclidean reading to the fourth decimal, and neither is detected as more hierarchical than a matched star, with MERU never ahead of its twin. Its embeddings never leave the near-flat regime of the hyperboloid: the spatial norm times the square root of the curvature stays between 0.258 and 0.279 at the median. Lorentz and Euclidean pairwise distances agree at a ratio of 0.997, so the control tests the training objective rather than a curved geometry (Table 8).

#### 5.4 WHOSE TREE

**The naive map manufactures an island.** Under Euclidean distances with average linkage the DINOv2 family agrees with the supervised and contrastive block at an adjusted Rand index of 0.03 against 0.64 within the block. The island is a chaining artifact: DINOv2’s Euclidean dendrograms put almost every class in one cluster, so the index is forced to zero by construction, and DINO-B does not fall in the island. Figure 5a shows the naive map.

**Controlling the cut leaves a moderate gap.** Under the criterion-selected configuration, cosine distances with average linkage, the DINOv2 family agrees with the block at 0.38 against 0.48 within the block (Figure 5b).

**The gap holds under every admissible cut.** Under the other three admissible configurations the DINOv2 family agrees with the block at 0.13 to 0.17. Within the block the agreement is 0.50 to 0.55, so about a third of it is missing under most configurations.

**The gap depends on the measure.** In cophenetic correlation the gap stays at 0.48 against 0.78. In triplet agreement, which needs no cut, it closes to 0.77 against 0.76 under cosine-average and persists under every other configuration (Table 10).

**The self-supervised tree is angular.** DINOv2’s semantics concentrate in the angular component of its representation. For DINOv2-L, sibling triplets on CIFAR-100 are resolved at 0.91 against 0.71 under cosine and under Euclidean distance respectively. For DINOv2-G the two figures are 0.85 and 0.63. This angular tree is why the Euclidean dendrograms chain into the island and why the triplet gap closes only under cosine-average.

**The cosine census agrees with the Euclidean one.** Repeating the census on L2-normalized centroids with geodesic distances and the same Haar null, the verdict matches the Euclidean record in 65 of 72 cells, and DINOv2 on ImageNet is genuine under both readings. The cosine census is robustness, never the record (Table 4).

**WordNet alignment follows supervision and recipe.** The Spearman correlation between inter-centroid and WordNet distances on ImageNet is +0.57 to +0.59 for the contrastive VLMs. It is +0.49 to +0.53 for the supervised ViTs.

**DINOv2 is the least aligned and the most robust.** For DINOv2 the correlation is +0.18 to +0.22, with shuffle and Gaussian controls at zero. Supervision deepens the resemblance to the human taxonomy on average across the supervised backbones tested, and the DeiT-B case of Section 5.3 shows that the effect is recipe-dependent. Across the three families, form and content anti-correlate: the most aligned families carry the most fragile excess, and DINOv2, the least aligned, the most robust one (Table 1).

**The recovery is not WordNet circularity.** On DBpedia Classes, an ontology independent of WordNet, the three text embedders read a genuine excess of  $-0.017$  to  $-0.021$  with every replicate above the real value. Their sibling-triplet agreement there is 0.89, and no clustering configuration degenerates.

**The same reading holds on a four-level caption hierarchy.** On HierarCaps the text encoders resolve sibling triplets at 0.83 to 0.97 and order the levels by projection radius, with no class set involved (Table 11).

**The text census depends on recipe, scale and probe.** On the thousand ImageNet class names, 7 of 15 text models are genuine under the record. GPT-2 S is genuine at  $p = .020$  and flips under the supremum and under cosine, GPT-2 M is not genuine, and GPT-2 L and XL are genuine under every reading. Pythia is genuine at every scale tested, and so is OLMo-1B. The sentence embedders are absent on the class names and present on DBpedia, so the probe decides their verdict. Prompt template and extraction protocol move the fragile values but not the scale ordering (Table 12).

## 5.5 WHAT IS SHARED IS LOCAL AND ORDINAL

**Local agreement survives calibration.** Permutation-calibrated mutual-kNN agreement over the sixty-six model pairs is 0.464, and every pair is significant at the resolution of the permutation null. Calibrated centroid CKA is 0.577, and the Poincaré readout improves neither reading. With the cut-free measures of Section 5.4, this is the reading of Gröger et al. (2026) on geometry: shared neighborhoods, a model-specific metric, and what a practitioner can collect from them is the subject of the next section (Table 13).

## 6 IMPLICATIONS FOR HYPERBOLIC REPRESENTATION LEARNING

**The raw reading cannot select a curvature.** Khrulkov et al. (2020) set the curvature of the Poincaré ball from the measured relative hyperbolicity,

$$c = (0.144/\delta_{\text{rel}})^2, \quad \delta_{\text{rel}} = 2\delta/\text{diam}. \quad (7)$$

Evaluated on the Gaussian band of Table 3, Eq. 7 assigns structureless clouds a curvature from 0.48 at  $d=192$  to 2.5 at  $d=1536$ , so the rule tracks the backbone’s width rather than its hierarchy.

**The calibrated reading predicts the zero-cost gain.** Within each hierarchical dataset, the record excess predicts the nearest-centroid gain of the Poincaré readout on all four datasets and the few-shot gain on three, with confidence intervals that exclude zero and a family control that leaves the within-dataset relation in place. The depth verdict predicts no gain anywhere (Table 14).

**A zero-cost readout collects what is there.** Cosine collects the self-supervised structure. The Poincaré readout adds from +0.9 to +1.3 pp over cosine for the contrastive VLMs on prototype tasks, and its effect on the other families is inconsistent in sign. Nothing helps on flat labels, and sample-level tasks never benefit. Before imposing curvature, then, a practitioner should read the excess and its rank, the depth test, whose positive verdict certifies and whose negative verdict says nothing, and the cosine reading that says where the structure lives (Table 14).

## 7 CONCLUSION AND LIMITATIONS

**Conclusion.** Read correctly, foundation models organize classes into clustered structure that is occasionally hierarchical and moderately shared; they do not converge to one common tree, and their raw tree-likeness is not evidence for hyperbolic geometry. The premise of latent hyperbolicity is confounded, and its sample-level reading sits within null noise. What survives at the class level is clustered structure in most cells, hierarchy above the superclasses in a few backbones, no detected depth from imposing the geometry, and a moderately shared, supervision-dependent recovery of the human taxonomy, angular for the self-supervised family. Curvature should be measured rather than assumed, and a zero-cost readout collects what is there.

**Limitations.** (i) The excess certifies structure beyond second moments, star-like clustering included, and depth needs the separate test. (ii) The depth test is conservative and weak at ImageNet’s noise level, certifies hierarchy above its frame only, leaves hierarchy within the clusters untested, and has no trained positive control that settles it, as Table 9 shows. (iii) The census is class-centroid geometry on six datasets. (iv) Causal-LM values, not their scale ordering, are template-conditioned. (v) The objective–geometry link is correlational, resting chiefly on DINOv2’s scale range and on two leaf-label ViTs. (vi) Downstream scope is prototype-based metric tasks with modest gains. (vii) The descriptor  $\xi$  tracks norm dispersion rather than curvature (Table 15), and a causal LM’s class representation depends on the extraction protocol as much as on the model. (viii) The sample-level reading covers two datasets and one image budget.

### REPRODUCIBILITY STATEMENT

Every number in the paper is annotated in the source with the result file that produces it, all tables are generated by script from those files (Table 16), and the full pipeline with fixed seeds will be released in the anonymized repository (`TODO(author)`). Its `tool/` directory ships the instrument as one script that reproduces any cell of Table 1 from its centroid matrix, and its output on two cells is checked against the table at every freeze. Estimator settings are stated in Section 4 and the model panel in Table 5.

### ETHICS STATEMENT

This work analyzes publicly released models and standard public datasets, involves no human subjects and no personal data, and proposes an analysis methodology and a metric-selection heuristic. We do not foresee ethical risks beyond those already associated with the underlying models and datasets.

### AI USE STATEMENT

In this work, we used generative AI tools for four tasks: writing assistance; drafting text from the authors’ outlines and results; implementation and execution of the experiments under the authors’ direction, with simulated reviews used as methodological feedback; and retrieval of references. We have not used generative AI tools for research ideation, and the remaining tasks with required disclosure are not applicable to this work. We have reviewed all AI-assisted work: the authors verified all code, numbers and references, every number in the paper is regenerated by script from the released result files and checked by an automated sweep at every freeze, and all experimental designs, results and claims were checked and approved by the authors. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

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## A PROOFS

*Proof of Lemma 1.* Each pairing sum in Eq. 1 adds two distances of the space, each of which lies in  $[d_{\min}, d_{\max}]$ . Hence  $S_{\max} \leq 2d_{\max}$  and  $S_{\min} \geq 2d_{\min}$ , so the defect  $\frac{1}{2}(S_{\max} - S_{\min})$  is at most  $d_{\max} - d_{\min}$  for every quadruple, and so is its supremum  $\delta$ . Dividing by the diameter  $d_{\max}$  gives  $\delta_{\text{norm}} \leq 1 - d_{\min}/d_{\max}$ ; the same bound holds for the 99.9th percentile of the sampled defects, which never exceeds their supremum.  $\square$

*Proof of Corollary 1.* Let  $x_1, \dots, x_n$  be iid standard Gaussian in  $\mathbb{R}^d$  with  $n$  fixed. For any pair  $i \neq j$  the coordinates of  $x_i - x_j$  are iid with variance 2, so  $\|x_i - x_j\|^2/(2d) \rightarrow 1$  almost surely as  $d \rightarrow \infty$  by the law of large numbers, and the convergence holds jointly over the finitely many pairs. Therefore  $d_{\min}/d_{\max} \rightarrow 1$ , and by Lemma 1  $\delta_{\text{norm}} \rightarrow 0$ . The rate is the distance-concentration rate of Beyer et al. (1999) and Aggarwal et al. (2001): the relative contrast  $(d_{\max} - d_{\min})/d_{\min}$  vanishes as  $d^{-1/2}$ . For a cloud with covariance  $\Sigma$  the same argument runs with  $d$  replaced by the effective dimension  $(\text{tr } \Sigma)^2/\text{tr}(\Sigma^2)$ , which is the spectrum confound of the remark, and a maximum over a growing sample of quadruples is non-decreasing in the sample size, which is the statistic confound.  $\square$

## B ONE TABLE PER QUESTION

Every table is generated by script from the released per-cell result files, and no value is hand-transcribed; the tables are numbered in the order the main text first cites them, one per question, with their result files indexed in Table 16.

### B.1 IS THE READING ROBUST? QUADRUPLE BUDGET, NULL CONSTRUCTION, RESAMPLING AND CLASS COUNT

**Hierarchy depth, not class count.** Because  $C$  sets the quadruple population, we sweep ImageNet subsets  $C \in \{10, \dots, 1000\}$  in two compositions: random (spanning the full hierarchy) and WordNet-coherent (siblings, effectively *flat*). At fixed  $C$ , random subsets are more exploitable than coherent ones and carry a larger excess at small  $C$  (CLIP-L at every  $C$ , DINOv2-L up to  $C=100$ , DINOv2-G up to  $C=20$ ; beyond that the two compositions converge). Random subsets still score  $-0.08$  to  $-0.14$  at  $C=10$ , so MNIST’s flat verdict is not a class-count artifact, and DINOv2-G stays below its null at every  $C$  (excess  $-0.008$  at  $C=1000, r=200$ ), where the supremum had placed it at the null.

(a) Quadruple budget: the excess at  $10^4$  to  $2 \times 10^6$  sampled quadruples per seed, under the record (Haar null, p99.9, 200 replicates) and under the superseded supremum  $\times$  Gaussian protocol.

| model    | data | protocol                   | $10^4$               | $5 \times 10^4$                                          | $10^5$  | $5 \times 10^5$ | $10^6$  | $2 \times 10^6$ | drift ( $\geq 10^5$ )     | drift/s.d. |
|----------|------|----------------------------|----------------------|----------------------------------------------------------|---------|-----------------|---------|-----------------|---------------------------|------------|
| ViT-L    | IN   | record                     | -0.0161              | -0.0161                                                  | -0.0160 | -0.0157         | -0.0159 | -0.0159         | 0.0002                    | 0.07       |
|          |      | supremum, Gaussian         | -0.026               | -0.024                                                   | -0.027  | -0.030          | -0.034  | -0.035          | p99.9 @ $5 \times 10^5$ : | -0.018     |
| DINOv2-L | IN   | record                     | -0.0131              | -0.0136                                                  | -0.0137 | -0.0135         | -0.0136 | -0.0136         | 0.0002                    | 0.38       |
|          |      | supremum, Gaussian         | -0.027               | -0.019                                                   | -0.024  | -0.011          | -0.004  | -0.005          | p99.9 @ $5 \times 10^5$ : | -0.024     |
| CLIP-B   | IN   | record                     | -0.0063              | -0.0060                                                  | -0.0059 | -0.0065         | -0.0064 | -0.0064         | 0.0007                    | 0.13       |
|          |      | supremum, Gaussian         | -0.008               | -0.012                                                   | -0.023  | -0.025          | -0.025  | -0.031          | p99.9 @ $5 \times 10^5$ : | -0.007     |
| ViT-L    | C100 | record                     | -0.0208              | -0.0208                                                  | -0.0206 | -0.0205         | -0.0205 | -0.0205         | 0.0002                    | 0.03       |
|          |      | supremum, Gaussian         | -0.045               | -0.047                                                   | -0.047  | -0.049          | -0.052  | -0.053          | p99.9 @ $5 \times 10^5$ : | -0.033     |
| DINOv2-L | C100 | record                     | -0.0348              | -0.0348                                                  | -0.0348 | -0.0348         | -0.0347 | -0.0347         | 0.0001                    | 0.07       |
|          |      | supremum, Gaussian         | -0.071               | -0.055                                                   | -0.054  | -0.050          | -0.052  | -0.055          | p99.9 @ $5 \times 10^5$ : | -0.069     |
| CLIP-B   | C100 | record                     | -0.0186              | -0.0195                                                  | -0.0188 | -0.0195         | -0.0194 | -0.0194         | 0.0006                    | 0.10       |
|          |      | supremum, Gaussian         | -0.045               | -0.047                                                   | -0.050  | -0.049          | -0.052  | -0.051          | p99.9 @ $5 \times 10^5$ : | -0.030     |
| ViT-L    | DTD  | record                     | -0.0305              | -0.0308                                                  | -0.0312 | -0.0311         | -0.0313 | -0.0313         | 0.0002                    | 0.05       |
|          |      | supremum, Gaussian         | -0.064               | -0.070                                                   | -0.076  | -0.083          | -0.084  | -0.084          | p99.9 @ $5 \times 10^5$ : | -0.054     |
| DINOv2-L | DTD  | record                     | -0.0418              | -0.0419                                                  | -0.0421 | -0.0420         | -0.0422 | -0.0420         | 0.0002                    | 0.06       |
|          |      | supremum, Gaussian         | -0.068               | -0.072                                                   | -0.075  | -0.083          | -0.084  | -0.084          | p99.9 @ $5 \times 10^5$ : | -0.064     |
| CLIP-B   | DTD  | record                     | -0.0243              | -0.0235                                                  | -0.0242 | -0.0240         | -0.0243 | -0.0242         | 0.0003                    | 0.05       |
|          |      | supremum, Gaussian         | -0.048               | -0.055                                                   | -0.059  | -0.065          | -0.066  | -0.066          | p99.9 @ $5 \times 10^5$ : | -0.035     |
| ViT-L    | C100 | centroid bootstrap: excess | -0.0493, s.d. 0.0029 | (30 resamples of the per-class images, 500 images/class) |         |                 |         |                 |                           |            |
| DINOv2-L | C100 | centroid bootstrap: excess | -0.0493, s.d. 0.0022 | (30 resamples of the per-class images, 500 images/class) |         |                 |         |                 |                           |            |
| DINOv2-G | C100 | centroid bootstrap: excess | -0.0778, s.d. 0.0020 | (30 resamples of the per-class images, 500 images/class) |         |                 |         |                 |                           |            |
| CLIP-B   | C100 | centroid bootstrap: excess | -0.0476, s.d. 0.0017 | (30 resamples of the per-class images, 500 images/class) |         |                 |         |                 |                           |            |
| ViT-L    | DTD  | centroid bootstrap: excess | -0.0803, s.d. 0.0049 | (30 resamples of the per-class images, 80 images/class)  |         |                 |         |                 |                           |            |
| DINOv2-L | DTD  | centroid bootstrap: excess | -0.0878, s.d. 0.0048 | (30 resamples of the per-class images, 80 images/class)  |         |                 |         |                 |                           |            |
| DINOv2-G | DTD  | centroid bootstrap: excess | -0.0863, s.d. 0.0051 | (30 resamples of the per-class images, 80 images/class)  |         |                 |         |                 |                           |            |
| CLIP-B   | DTD  | centroid bootstrap: excess | -0.0581, s.d. 0.0054 | (30 resamples of the per-class images, 80 images/class)  |         |                 |         |                 |                           |            |

(b) ImageNet centroid bootstrap under the record: 30 resamples of the 100 training images per class (20 null replicates per resample).

| model    | excess (reference) | bootstrap mean | bootstrap s.d. | fraction negative |
|----------|--------------------|----------------|----------------|-------------------|
| ViT-T    | -0.0055            | -0.0051        | 0.0011         | 1.00              |
| ViT-S    | -0.0126            | -0.0122        | 0.0007         | 1.00              |
| ViT-B    | -0.0165            | -0.0160        | 0.0006         | 1.00              |
| ViT-L    | -0.0160            | -0.0160        | 0.0004         | 1.00              |
| DINO-B   | -0.0209            | -0.0208        | 0.0003         | 1.00              |
| DINOv2-S | -0.0164            | -0.0162        | 0.0004         | 1.00              |
| DINOv2-B | -0.0167            | -0.0164        | 0.0002         | 1.00              |
| DINOv2-L | -0.0136            | -0.0134        | 0.0001         | 1.00              |
| DINOv2-G | -0.0080            | -0.0083        | 0.0004         | 1.00              |
| CLIP-B   | -0.0068            | -0.0069        | 0.0009         | 1.00              |
| CLIP-L   | -0.0094            | -0.0095        | 0.0006         | 1.00              |
| SigLIP-B | -0.0080            | -0.0081        | 0.0006         | 1.00              |

Table 2: **The excess is budget-stable, its size is partly a property of the null construction, and the genuine count survives resampling and estimator noise.** (a) Excess against the quadruple budget for nine cells; drift = range of the record excess over budgets  $\geq 10^5$ , divided by the excess’s spread in the last column. Under the record every sign holds at every budget and the drift of the excess from  $10^5$  upward is at most 0.38 of the excess’s own spread (null and estimator s.d. combined at  $5 \times 10^5$ ) on ImageNet, 0.10 on CIFAR-100 and 0.06 on DTD, so the excess is budget-stable; the supremum rows show the drift of up to 0.02 on ImageNet that the superseded protocol carried. (b) Bootstrap of the ImageNet excess over resampled images per class: the bootstrap s.d. is at most 0.0011 and every resample of every backbone is sign-negative.

(c) Three constructions of the spectrum-matched null (10 replicates each): Gaussian coefficients, Haar-rotated coefficients (exact sample spectrum), PC-permutation (exact per-component marginals).

| model    | IN     |        |          | C100   |        |          | C10    |        |          |
|----------|--------|--------|----------|--------|--------|----------|--------|--------|----------|
|          | Gauss. | Haar   | PC-perm. | Gauss. | Haar   | PC-perm. | Gauss. | Haar   | PC-perm. |
| ViT-T    | -0.005 | -0.014 | -0.038   | -0.044 | -0.031 | -0.055   | -0.057 | -0.027 | -0.050   |
| ViT-S    | -0.023 | -0.025 | -0.052   | -0.034 | -0.025 | -0.049   | -0.098 | -0.063 | -0.090   |
| ViT-B    | -0.027 | -0.025 | -0.054   | -0.018 | -0.011 | -0.039   | -0.090 | -0.054 | -0.078   |
| ViT-L    | -0.028 | -0.029 | -0.040   | -0.047 | -0.034 | -0.051   | -0.104 | -0.066 | -0.096   |
| DINO-B   | -0.005 | -0.002 | -0.008   | -0.068 | -0.056 | -0.068   | -0.100 | -0.070 | -0.097   |
| DINOv2-S | +0.010 | +0.015 | +0.013   | -0.050 | -0.029 | -0.051   | -0.093 | -0.083 | -0.098   |
| DINOv2-B | +0.004 | +0.015 | +0.006   | -0.042 | -0.007 | -0.041   | -0.132 | -0.120 | -0.124   |
| DINOv2-L | -0.013 | +0.000 | -0.011   | -0.053 | -0.008 | -0.044   | -0.153 | -0.136 | -0.122   |
| DINOv2-G | -0.000 | +0.016 | +0.001   | -0.080 | -0.032 | -0.074   | -0.155 | -0.138 | -0.110   |
| CLIP-B   | -0.028 | -0.028 | -0.048   | -0.046 | -0.036 | -0.053   | -0.079 | -0.045 | -0.075   |
| CLIP-L   | -0.016 | -0.017 | -0.029   | -0.042 | -0.031 | -0.052   | -0.087 | -0.049 | -0.082   |
| SigLIP-B | -0.019 | -0.020 | -0.044   | -0.042 | -0.032 | -0.046   | -0.096 | -0.056 | -0.081   |

(d) Joint sensitivity of the genuine count:  $z_{\text{joint}}$  against null, bootstrap and estimator noise, and (in parentheses) the number of the 30 centroid resamples in which the cell is genuine.

| model    | IN                     | C100                  | C10                    | DTD                    | FMNIST                  | MNIST                  |
|----------|------------------------|-----------------------|------------------------|------------------------|-------------------------|------------------------|
| ViT-T    | -1.1 (0) <sup>o</sup>  | -0.7 (0) <sup>o</sup> | -1.1 (0) <sup>o</sup>  | -2.0 (13) <sup>†</sup> | +0.9 (0) <sup>o</sup>   | -1.0 (0) <sup>o</sup>  |
| ViT-S    | -3.6 (30)              | -0.0 (0) <sup>o</sup> | -2.7 (30)              | -3.1 (30)              | -1.7 (30) <sup>†</sup>  | -1.3 (0) <sup>o</sup>  |
| ViT-B    | -5.2 (30)              | +0.8 (0) <sup>o</sup> | -2.4 (30)              | -4.0 (30)              | -2.1 (30)               | -0.6 (0) <sup>o</sup>  |
| ViT-L    | -5.0 (30)              | -4.0 (30)             | -2.8 (30)              | -6.4 (30)              | -2.5 (30)               | -1.6 (0) <sup>o</sup>  |
| DINO-B   | -8.9 (30)              | -7.6 (30)             | -2.7 (30)              | -4.5 (30)              | -0.3 (0) <sup>o</sup>   | -1.5 (0) <sup>o</sup>  |
| DINOv2-S | -15.9 (30)             | -9.7 (30)             | -2.8 (30)              | -3.2 (30)              | -1.6 (30) <sup>o‡</sup> | -0.2 (0) <sup>o</sup>  |
| DINOv2-B | -31.6 (30)             | -14.7 (30)            | -4.3 (30)              | -8.1 (30)              | -2.7 (30)               | -0.6 (0) <sup>o</sup>  |
| DINOv2-L | -30.7 (30)             | -21.6 (30)            | -4.9 (30)              | -10.6 (30)             | -1.8 (29) <sup>†</sup>  | -0.5 (0) <sup>o</sup>  |
| DINOv2-G | -11.5 (30)             | -28.5 (30)            | -4.8 (30)              | -8.2 (30)              | -2.0 (30)               | -0.8 (0) <sup>o</sup>  |
| CLIP-B   | -1.3 (0) <sup>o</sup>  | -3.1 (30)             | -1.8 (7) <sup>†</sup>  | -3.5 (30)              | -1.3 (0) <sup>o</sup>   | -0.0 (0) <sup>o</sup>  |
| CLIP-L   | -2.1 (12) <sup>†</sup> | -3.5 (30)             | -2.0 (30) <sup>†</sup> | -4.3 (30)              | -1.5 (0) <sup>o</sup>   | -0.4 (0) <sup>o</sup>  |
| SigLIP-B | -1.7 (0) <sup>o</sup>  | -4.3 (30)             | -2.6 (30)              | -4.1 (30)              | -2.0 (30) <sup>†</sup>  | -1.9 (30) <sup>†</sup> |

Table 2: (continued) (c) Excess under three constructions of the spectrum null: Gaussian and PC-permutation agree, the exact-sample-spectrum construction yields smaller excesses for low-effective-rank clouds on the small- $C$  sets (most visibly DINOv2 on CIFAR-100), while signs and the ImageNet readings are stable. (d) Per cell:  $z_{\text{joint}} = \text{excess} / \sqrt{\sigma_{\text{null}}^2 + \sigma_{\text{boot}}^2 + \sigma_{\text{est}}^2}$  (200-replicate null s.d., bootstrap s.d. over 30 centroid resamples, quadruple-seed s.d.) and the number of resamples in which the cell is genuine under BH over the 72 cells (50 Haar replicates per resample); <sup>o</sup>: not genuine in the record; <sup>†</sup>: record-genuine but failing one of the two criteria ( $z_{\text{joint}} \leq -2$ , genuine in  $\geq 27$  of 30); <sup>‡</sup>: not record-genuine but passing one. Counts: record 49/72 (18/24);  $z_{\text{joint}} \leq -2$ : 42/72 (18/24); genuine in  $\geq 27$  of 30 resamples: 47/72 (17/24).

(e) Class count against hierarchy depth: ImageNet subsets of  $C$  classes, random (spanning the hierarchy) or WordNet-coherent (siblings, effectively flat), under the record.

| model    | $C$  | random subsets (span the hierarchy) |        |       |        | WordNet-coherent (siblings) |        |       |        |
|----------|------|-------------------------------------|--------|-------|--------|-----------------------------|--------|-------|--------|
|          |      | $\hat{\delta}_{99.9}$               | excess | below | NC adv | $\hat{\delta}_{99.9}$       | excess | below | NC adv |
| DINOv2-L | 10   | 0.030                               | -0.134 | 5/5   | +1.04  | 0.068                       | -0.098 | 5/5   | +0.08  |
| DINOv2-L | 20   | 0.034                               | -0.091 | 5/5   | +1.32  | 0.071                       | -0.065 | 5/5   | +0.12  |
| DINOv2-L | 50   | 0.032                               | -0.046 | 5/5   | +1.58  | 0.050                       | -0.042 | 5/5   | +0.53  |
| DINOv2-L | 100  | 0.031                               | -0.032 | 5/5   | +1.46  | 0.044                       | -0.031 | 5/5   | +0.77  |
| DINOv2-L | 200  | 0.031                               | -0.022 | 5/5   | +1.45  | 0.041                       | -0.023 | 5/5   | +0.59  |
| DINOv2-L | 500  | 0.030                               | -0.016 | 2/2   | +1.01  | 0.031                       | -0.018 | 2/2   | +0.82  |
| DINOv2-L | 1000 | 0.030                               | -0.014 | 1/1   | +0.57  | —                           | —      | —     | —      |
| DINOv2-G | 10   | 0.029                               | -0.136 | 5/5   | +0.80  | 0.072                       | -0.095 | 5/5   | -0.56  |
| DINOv2-G | 20   | 0.037                               | -0.089 | 5/5   | +1.30  | 0.070                       | -0.067 | 5/5   | -0.18  |
| DINOv2-G | 50   | 0.039                               | -0.039 | 5/5   | +1.51  | 0.053                       | -0.042 | 5/5   | +0.26  |
| DINOv2-G | 100  | 0.041                               | -0.022 | 5/5   | +1.30  | 0.049                       | -0.028 | 5/5   | +0.36  |
| DINOv2-G | 200  | 0.036                               | -0.015 | 5/5   | +1.28  | 0.045                       | -0.020 | 5/5   | +0.27  |
| DINOv2-G | 500  | 0.034                               | -0.009 | 2/2   | +0.49  | 0.034                       | -0.013 | 2/2   | +0.22  |
| DINOv2-G | 1000 | 0.033                               | -0.008 | 1/1   | -0.13  | —                           | —      | —     | —      |
| CLIP-L   | 10   | 0.071                               | -0.080 | 5/5   | -0.08  | 0.099                       | -0.066 | 4/5   | -0.36  |
| CLIP-L   | 20   | 0.078                               | -0.047 | 5/5   | +0.20  | 0.113                       | -0.036 | 4/5   | -0.24  |
| CLIP-L   | 50   | 0.080                               | -0.027 | 5/5   | +0.34  | 0.113                       | -0.014 | 4/5   | -0.46  |
| CLIP-L   | 100  | 0.079                               | -0.024 | 5/5   | +0.21  | 0.095                       | -0.016 | 3/5   | -0.31  |
| CLIP-L   | 200  | 0.078                               | -0.019 | 5/5   | +0.00  | 0.086                       | -0.010 | 2/5   | -0.35  |
| CLIP-L   | 500  | 0.077                               | -0.012 | 2/2   | -0.27  | 0.081                       | -0.011 | 1/2   | -0.42  |
| CLIP-L   | 1000 | 0.076                               | -0.009 | 1/1   | -0.53  | —                           | —      | —     | —      |

Table 2: (continued) (e) Class-count control under the record (means over subset seeds: 5 for  $C \leq 200$ , 2 for  $C=500$ , 1 for  $C=1000$ ); “below” = subset seeds whose real value lies below the null at uncorrected  $p \leq 0.05$ ; NC adv = prototype-classifier advantage of the Poincaré readout in pp.

## B.2 IS A LOW RAW READING EVIDENCE? CALIBRATION ON REFERENCE GEOMETRIES AND SYNTHETIC CLUSTERS

| (a) Reference geometries.                               |                     |                                     |  |  |  |
|---------------------------------------------------------|---------------------|-------------------------------------|--|--|--|
| space                                                   | absolute $\delta$   | $\delta_{\text{norm}}$              |  |  |  |
| balanced binary tree (depth 10)                         | 0.000               | 0.000                               |  |  |  |
| $\mathbb{H}^2$ ( $K = -1$ ), region radius $R=2/4/8/16$ | 0.65/0.69/0.69/0.69 | 0.162/0.087/0.043/0.022             |  |  |  |
| uniform $S^{99}$ (chord / geodesic)                     | 0.25/0.37           | 0.143/0.179                         |  |  |  |
| iid Gaussian ( $n=1000$ ), $d=192, \dots, 1536$         | —                   | 0.104/0.081/0.074/0.061/0.055/0.046 |  |  |  |

| (b) Synthetic clusters: what each null certifies. |                |                 |        |                        |        |
|---------------------------------------------------|----------------|-----------------|--------|------------------------|--------|
| synthetic cloud ( $n=1000$ , $d=768$ )            | $\hat{\delta}$ | spectrum null A |        | hub-randomizing null B |        |
|                                                   |                | Gaussian        | Haar   | Gaussian               | Haar   |
| 30-cluster star, tight (0.1)                      | 0.055          | −0.115          | −0.115 | −0.126                 | −0.058 |
| 30-cluster star, mid (0.3)                        | 0.059          | −0.104          | −0.103 | −0.114                 | −0.049 |
| 30-cluster star, loose (0.6)                      | 0.061          | −0.084          | −0.082 | −0.090                 | −0.033 |
| 2-level hierarchy $6 \times 5$ (0.3)              | 0.044          | −0.143          | −0.143 | −0.113                 | −0.112 |

Table 3: **A low raw  $\delta$  is not evidence of hierarchy, and the spectrum excess certifies clustering, not depth.** (a) Calibration on reference geometries: the estimator recovers the tree zero and the four-point constant of  $\mathbb{H}^2$ , spheres stay high, and iid Gaussians drift low as  $d$  grows, the dimension confound; the ordering trees  $<$  hyperbolic  $<$  spherical holds for hyperbolic regions of radius  $R \geq 4$ . Gaussian row from the released control file; tree/ $\mathbb{H}^2$ /sphere rows re-verified synthetically at every freeze. (b) Excess of synthetic clouds (means over seeds; within/between noise ratio in parentheses) under the spectrum null A, with Gaussian coefficients and with Haar-rotated coefficients (exact sample spectrum), and under the hub-randomizing null B, which keeps every cluster and resamples the hub configuration. A pure star sits far below null A under either construction, so null A certifies clustering, not depth; null B also reports the star as hierarchical, since a near-regular simplex of hubs already minimizes  $\delta$ , which is why the depth test of Table 8 is read against a matched star rather than against null B. Under the Haar construction the two-level hierarchy sits 2–3 $\times$  below the star; under the plain Gaussian null the mid-radius star scores −0.104 and the  $6 \times 5$  hierarchy −0.14, a fraction of either’s excess.

## B.3 IS THE CLUSTERED STRUCTURE GENUINE? THE VISION CENSUS UNDER EVERY CONSTRUCTION

**What each null preserves.** The spectrum-matched null, Gaussian or Haar, preserves  $n$ ,  $d$ , the mean and the covariance spectrum and destroys all higher-order structure; the PC-permutation null preserves each principal component’s marginal and destroys the dependence across components; the label shuffle keeps the point cloud itself and destroys the embedding–label correspondence; and the hub-randomizing null keeps each cluster’s geometry and hub assignment and destroys the hub arrangement, which is why it serves the depth test only when read against a matched star. “Genuine” in the text always means beyond the first of these. In plain terms, the spectrum null is a cloud with the same shape as the real one and no structure inside it; the hub null of Section 5 keeps every cluster of the real cloud and only rearranges where the clusters sit.

(a) The census of record per cell: Haar spectrum-matched null, 99.9th-percentile statistic, 200 replicates.

| model    | IN      |     |      | C100    |     |      | C10     |     |      | DTD    |     |      | FMNIST  |     |      | MNIST   |     |      |
|----------|---------|-----|------|---------|-----|------|---------|-----|------|--------|-----|------|---------|-----|------|---------|-----|------|
|          | exc.    | r   | p    | exc.    | r   | p    | exc.    | r   | p    | exc.   | r   | p    | exc.    | r   | p    | exc.    | r   | p    |
| ViT-T    | -0.005° | 169 | .159 | -0.006° | 151 | .249 | -0.021° | 172 | .144 | -0.014 | 197 | .020 | +0.018° | 34  | .831 | -0.026° | 168 | .164 |
| ViT-S    | -0.012  | 200 | .005 | -0.000° | 111 | .448 | -0.056  | 200 | .005 | -0.019 | 199 | .010 | -0.033  | 196 | .025 | -0.037° | 181 | .100 |
| ViT-B    | -0.016  | 200 | .005 | +0.006° | 43  | .786 | -0.050  | 200 | .005 | -0.021 | 200 | .005 | -0.042  | 200 | .005 | -0.013° | 142 | .294 |
| ViT-L    | -0.016  | 200 | .005 | -0.020  | 200 | .005 | -0.060  | 200 | .005 | -0.031 | 200 | .005 | -0.051  | 200 | .005 | -0.040° | 193 | .040 |
| DINO-B   | -0.020  | 200 | .005 | -0.032  | 200 | .005 | -0.061  | 200 | .005 | -0.027 | 200 | .005 | -0.007° | 127 | .368 | -0.034° | 188 | .065 |
| DINOv2-S | -0.016  | 200 | .005 | -0.031  | 200 | .005 | -0.071  | 200 | .005 | -0.018 | 200 | .005 | -0.031° | 193 | .040 | -0.004° | 112 | .443 |
| DINOv2-B | -0.017  | 200 | .005 | -0.031  | 200 | .005 | -0.112  | 200 | .005 | -0.035 | 200 | .005 | -0.062  | 200 | .005 | -0.017° | 147 | .269 |
| DINOv2-L | -0.014  | 200 | .005 | -0.035  | 200 | .005 | -0.128  | 200 | .005 | -0.042 | 200 | .005 | -0.035  | 197 | .020 | -0.014° | 132 | .343 |
| DINOv2-G | -0.008  | 200 | .005 | -0.040  | 200 | .005 | -0.130  | 200 | .005 | -0.039 | 200 | .005 | -0.046  | 199 | .010 | -0.022° | 158 | .214 |
| CLIP-B   | -0.007° | 172 | .144 | -0.019  | 200 | .005 | -0.036  | 195 | .030 | -0.024 | 200 | .005 | -0.035° | 181 | .100 | -0.001° | 104 | .483 |
| CLIP-L   | -0.009  | 197 | .020 | -0.018  | 200 | .005 | -0.043  | 199 | .010 | -0.027 | 200 | .005 | -0.037° | 188 | .065 | -0.008° | 126 | .373 |
| SigLIP-B | -0.008° | 189 | .060 | -0.022  | 200 | .005 | -0.050  | 199 | .010 | -0.023 | 200 | .005 | -0.042  | 198 | .015 | -0.044  | 198 | .015 |

(b) Verdict under the four null  $\times$  statistic constructions and the excess as a fraction of the null reading.

| model    | IN       | C100     | C10      | DTD      | FMNIST   | MNIST    | sup. IN exc. (r) |
|----------|----------|----------|----------|----------|----------|----------|------------------|
| ViT-T    | ●●● -5%  | ●●● -5%  | ○○○ -15% | ●●● -11% | ○○○ +14% | ○○○ -16% | -0.020 (199)     |
| ViT-S    | ●●● -16% | ●●● -0%  | ●●● -39% | ●●● -16% | ●●● -25% | ○○○ -22% | -0.020 (200)     |
| ViT-B    | ●●● -23% | ○○○ +5%  | ●●● -36% | ●●● -19% | ●●● -30% | ○○○ -9%  | -0.022 (200)     |
| ViT-L    | ●●● -22% | ●●● -18% | ●●● -42% | ●●● -29% | ●●● -35% | ○○○ -24% | -0.027 (200)     |
| DINO-B   | ●●○ -27% | ●●● -28% | ●●● -41% | ●●● -22% | ○○○ -5%  | ○○○ -24% | +0.000 (96)      |
| DINOv2-S | ●●○ -23% | ●●● -31% | ●●● -46% | ●●● -15% | ○○● -23% | ○○○ -3%  | +0.017 (0)       |
| DINOv2-B | ●●○ -33% | ●●● -39% | ●●● -71% | ●●● -31% | ●●● -43% | ○○○ -11% | +0.018 (0)       |
| DINOv2-L | ●●● -31% | ●●● -49% | ●●● -82% | ●●● -38% | ●●● -25% | ○○○ -8%  | +0.006 (0)       |
| DINOv2-G | ●●○ -20% | ●●● -57% | ●●● -84% | ●●● -35% | ●●● -31% | ○○○ -13% | +0.020 (0)       |
| CLIP-B   | ●●○ -7%  | ●●● -16% | ●●● -26% | ●●● -20% | ○○○ -20% | ○○○ -1%  | -0.023 (199)     |
| CLIP-L   | ●●● -11% | ●●● -16% | ●●● -30% | ●●● -21% | ○○○ -24% | ○○○ -5%  | -0.016 (198)     |
| SigLIP-B | ○○● -9%  | ●●● -18% | ●●● -36% | ●●● -20% | ●●● -27% | ●●● -28% | -0.015 (194)     |

Table 4: **Clustered structure is the rule at the class level under every construction of the null and statistic, and the verdict does not depend on the metric.** (a) Per cell: excess of  $\hat{\delta}_{99.9}$  over the null mean,  $r$  = replicates above the real value,  $p = (1 + \#\{\text{null} \leq \text{real}\})/201$ ; genuine = Benjamini-Hochberg-corrected  $p \leq 0.05$  over the 72 cells (°: not genuine). Genuine cells: 49/72 overall, 18/24 on ImageNet+CIFAR-100; sign agreement with the original 5-replicate Gaussian  $\times$  supremum census on the 72 comparable cells: 67/72. (b) Four symbols per cell in the order Haar  $\times$  p99.9 (the record), Haar  $\times$  supremum, Gaussian  $\times$  p99.9, Gaussian  $\times$  supremum, ● = genuine under that construction; genuine counts 49/72, 46/72, 52/72, 52/72. The percentage is the record excess divided by the mean null reading (ImageNet -33% to -5%; all cells -84% to +14%); the last column is the ImageNet excess of the supremum statistic under the same Haar null with its rank. The Haar construction reproduces the sample spectrum exactly; the Gaussian one does not at small  $n$  and inflates excesses for low-rank clouds (Table 2). The supremum is decided by a few extreme quadruples and disagrees with the 99.9th percentile in both directions on DINOv2-S/B/G ImageNet.

(c) The same census on cosine geometry (L2-normalized centroids, geodesic distances, Haar null on the normalized cloud): robustness, not the record.

| model    | IN     |     | C100                |     | C10    |     | DTD    |     | FMNIST              |     | MNIST               |     |
|----------|--------|-----|---------------------|-----|--------|-----|--------|-----|---------------------|-----|---------------------|-----|
|          | exc.   | $r$ | exc.                | $r$ | exc.   | $r$ | exc.   | $r$ | exc.                | $r$ | exc.                | $r$ |
| ViT-T    | -0.012 | 200 | -0.014 <sup>o</sup> | 191 | -0.043 | 198 | -0.017 | 199 | +0.007 <sup>o</sup> | 73  | -0.026 <sup>o</sup> | 167 |
| ViT-S    | -0.017 | 200 | -0.017              | 200 | -0.073 | 200 | -0.021 | 200 | -0.049              | 200 | -0.036 <sup>o</sup> | 183 |
| ViT-B    | -0.022 | 200 | -0.006 <sup>o</sup> | 170 | -0.066 | 200 | -0.021 | 200 | -0.041              | 199 | -0.010 <sup>o</sup> | 130 |
| ViT-L    | -0.021 | 200 | -0.024              | 200 | -0.080 | 200 | -0.024 | 200 | -0.066              | 200 | -0.043              | 196 |
| DINO-B   | -0.021 | 200 | -0.035              | 200 | -0.070 | 200 | -0.024 | 200 | -0.006 <sup>o</sup> | 116 | -0.038 <sup>o</sup> | 193 |
| DINOv2-S | -0.014 | 200 | -0.033              | 200 | -0.084 | 200 | -0.018 | 200 | -0.054              | 200 | -0.004 <sup>o</sup> | 113 |
| DINOv2-B | -0.013 | 200 | -0.032              | 200 | -0.140 | 200 | -0.021 | 200 | -0.081              | 200 | -0.019 <sup>o</sup> | 147 |
| DINOv2-L | -0.010 | 200 | -0.029              | 200 | -0.151 | 200 | -0.024 | 200 | -0.051              | 200 | -0.020 <sup>o</sup> | 145 |
| DINOv2-G | -0.006 | 200 | -0.029              | 200 | -0.169 | 200 | -0.017 | 200 | -0.064              | 200 | -0.023 <sup>o</sup> | 157 |
| CLIP-B   | -0.017 | 200 | -0.023              | 200 | -0.039 | 197 | -0.031 | 200 | -0.038 <sup>o</sup> | 185 | +0.002 <sup>o</sup> | 93  |
| CLIP-L   | -0.017 | 200 | -0.025              | 200 | -0.045 | 199 | -0.031 | 200 | -0.038 <sup>o</sup> | 192 | -0.008 <sup>o</sup> | 124 |
| SigLIP-B | -0.013 | 200 | -0.025              | 200 | -0.053 | 200 | -0.027 | 200 | -0.052              | 199 | -0.044              | 197 |

(d) Backbones outside the ViT panel on the ImageNet centroid store (spectrum null, 20 replicates).

| model                    | $d$  | $\hat{\delta}$ | excess ( $z$ ) | reps above |
|--------------------------|------|----------------|----------------|------------|
| Barlow Twins (ResNet-50) | 2048 | 0.075          | -0.039 (-4.8)  | 20/20      |
| BYOL (ResNet-50)         | 2048 | 0.073          | -0.020 (-2.4)  | 20/20      |

Table 4: (continued) (c) Cosine reading, BH over 72 cells: sign-negative 70/72, genuine 56/72 and 22/24 on ImageNet+CIFAR-100, verdict agreement with the Euclidean record 65/72 cells. Text under the same cosine protocol (padding-free embeddings): genuine 8/15: gpt2-m, gpt2-l, gpt2-xl, pythia-410m, pythia-1b, pythia-2b8, olmo-1b, gte-qwen2. The angular tree: sibling-triplet agreement on CIFAR-100 for DINOv2-L is 0.91 under cosine against 0.71 under Euclidean distance. (d) Two self-supervised ResNet-50 backbones read the same clustered structure as the ViTs.

## B.4 THE MODEL PANEL AND EXTRACTION

| type                    | model                              | params       |
|-------------------------|------------------------------------|--------------|
| Supervised ViT (vision) | ViT-T (i21k)                       | 5M           |
| Supervised ViT (vision) | ViT-S (i21k)                       | 22M          |
| Supervised ViT (vision) | ViT-B (i21k)                       | 86M          |
| Supervised ViT (vision) | ViT-L (i21k)                       | 307M         |
| SSL (vision)            | DINO-B                             | 86M          |
| SSL (vision)            | DINOv2-S                           | 22M          |
| SSL (vision)            | DINOv2-B                           | 86M          |
| SSL (vision)            | DINOv2-L                           | 307M         |
| SSL (vision)            | DINOv2-G                           | 1.1B         |
| Contrastive (vision)    | CLIP-B                             | 86M          |
| Contrastive (vision)    | CLIP-L                             | 307M         |
| Contrastive (vision)    | SigLIP-B                           | 86M          |
| Causal LM               | GPT-2 S                            | 117M         |
| Causal LM               | GPT-2 M                            | 345M         |
| Causal LM               | GPT-2 L                            | 774M         |
| Causal LM               | GPT-2 XL                           | 1.5B         |
| Causal LM               | Pythia-410M                        | 410M         |
| Causal LM               | Pythia-1B                          | 1.0B         |
| Causal LM               | Pythia-2.8B                        | 2.8B         |
| Causal LM               | OLMo-1B                            | 1.0B         |
| Causal LM               | OLMo-7B                            | 7.0B         |
| Text embedder           | BGE-base                           | 110M         |
| Text embedder           | BGE-large                          | 335M         |
| Text embedder           | GTE-base                           | 110M         |
| Text embedder           | GTE-large                          | 335M         |
| Text embedder           | GTE-Qwen2-1.5B                     | 1.5B         |
| Text embedder           | E5-base                            | 110M         |
| Text embedder           | E5-large                           | 335M         |
| Controls (vision)       | DeiT-B/16 (IN-1k, no distillation) | 86M          |
| Controls (vision)       | ViT-B/16 augreg (IN-1k)            | 86M          |
| Controls (vision)       | MERU ViT-S/B/L and CLIP twins      | 22M/86M/307M |
| Controls (vision)       | Barlow Twins, BYOL (ResNet-50)     | 24M          |

Table 5: **The model panel.** 12 vision backbones (all in the census; the 10 with cached per-dataset features, all but DINO-B and SigLIP-B, form the task grid), 9 causal LMs, 7 text embedders, and the vision models used only as controls. Class centroids average 100 cached training images per class on ImageNet and the full cached split, 80–6,000 images per class, on the other datasets. Feature extraction: vision backbones use the encoder’s default pooled embedding (timm, num\_classes=0, CLS or global pool per architecture); causal LMs use the last-token hidden state, extracted one prompt at a time (padding-free: left-padded batching alters GPT-2’s hidden states, Table 12); text embedders use their recommended pooling (CLS for BGE, mean for GTE/E5, last token for GTE-Qwen2); CLIP/SigLIP text towers use their text encoders.

## B.5 DOES THE PREMISE SURVIVE AT THE SAMPLE LEVEL? THE CENSUS ON PER-IMAGE FEATURES

| model    | CIFAR-100 (10 images/class, $n=1000$ ) |                       |                 |           |                 | DTD (22 images/class, $n=1034$ ) |                       |                 |           |                 |
|----------|----------------------------------------|-----------------------|-----------------|-----------|-----------------|----------------------------------|-----------------------|-----------------|-----------|-----------------|
|          | sup. $\hat{\delta}$                    | $\hat{\delta}_{99.9}$ | excess          | exc./null | $r/200$ ( $p$ ) | sup. $\hat{\delta}$              | $\hat{\delta}_{99.9}$ | excess          | exc./null | $r/200$ ( $p$ ) |
| ViT-T    | 0.130                                  | 0.093                 | $-0.0016^\circ$ | -2%       | 134 (0.333)     | 0.124                            | 0.087                 | $-0.0004^\circ$ | -0%       | 111 (0.448)     |
| ViT-S    | 0.121                                  | 0.086                 | $+0.0051^\circ$ | +6%       | 21 (0.896)      | 0.103                            | 0.070                 | $-0.0062^\circ$ | -8%       | 197 (0.020)     |
| ViT-B    | 0.121                                  | 0.082                 | $+0.0031^\circ$ | +4%       | 50 (0.751)      | 0.092                            | 0.060                 | $-0.0080^\circ$ | -12%      | 200 (0.005)     |
| ViT-L    | 0.100                                  | 0.067                 | $-0.0041^\circ$ | -6%       | 185 (0.080)     | 0.092                            | 0.056                 | $-0.0073^\circ$ | -12%      | 200 (0.005)     |
| DINO-B   | 0.104                                  | 0.070                 | $-0.0042^\circ$ | -6%       | 194 (0.035)     | 0.110                            | 0.075                 | $+0.0008^\circ$ | +1%       | 80 (0.602)      |
| DINOv2-S | 0.100                                  | 0.065                 | $-0.0049^\circ$ | -7%       | 200 (0.005)     | 0.145                            | 0.090                 | $+0.0058^\circ$ | +7%       | 0 (1.000)       |
| DINOv2-B | 0.089                                  | 0.050                 | $-0.0090^\circ$ | -15%      | 200 (0.005)     | 0.139                            | 0.073                 | $-0.0019^\circ$ | -3%       | 162 (0.194)     |
| DINOv2-L | 0.087                                  | 0.043                 | $-0.0131^\circ$ | -24%      | 200 (0.005)     | 0.131                            | 0.064                 | $-0.0060^\circ$ | -9%       | 200 (0.005)     |
| DINOv2-G | 0.102                                  | 0.036                 | $-0.0232^\circ$ | -39%      | 200 (0.005)     | 0.136                            | 0.065                 | $-0.0079^\circ$ | -11%      | 200 (0.005)     |
| CLIP-B   | 0.117                                  | 0.080                 | $-0.0060^\circ$ | -7%       | 184 (0.085)     | 0.125                            | 0.089                 | $+0.0021^\circ$ | +2%       | 71 (0.647)      |
| SigLIP-B | 0.113                                  | 0.078                 | $-0.0074^\circ$ | -9%       | 199 (0.010)     | 0.120                            | 0.082                 | $-0.0008^\circ$ | -1%       | 124 (0.383)     |
|          |                                        |                       |                 |           |                 | 0.114                            | 0.079                 | $-0.0044^\circ$ | -5%       | 184 (0.085)     |

Table 6: **The sample-level reading, the object of prior latent-hyperbolicity claims, sits within null noise in most cells.** The instrument on  $\approx 1000$  stratified training images per cell (per-image features, not centroids): sup.  $\hat{\delta}$  is the raw supremum  $\delta_{\text{norm}}$  the literature reports;  $\hat{\delta}_{99.9}$ , excess, exc./null,  $r/200$  and  $p$  follow Table 1 (Haar spectrum-matched null on the sample cloud, 99.9th-percentile statistic, 200 replicates);  $^\circ$ : not genuine under Benjamini–Hochberg over the 24 cells. The raw supremum lies in the band 0.087–0.145; the excess is not genuine in 14 of 24 cells and genuine in 10 (DINOv2-S, DINOv2-B, DINOv2-L, DINOv2-G, SigLIP-B on CIFAR-100; ViT-S, ViT-B, ViT-L, DINOv2-L, DINOv2-G on DTD), where it reaches at most  $-0.023$ , that is  $-39\%$  to  $-7\%$  of the null reading.

## B.6 IS THE STRUCTURE LEARNED? TRAINING INTERVENTIONS AND CURVATURE BY EDGE TYPE

**Clustered structure tracks training.** Two interventions move it (Figure 6; raw  $\delta$ , within architecture): fine-tuning on a task with no class hierarchy raises  $\delta$  by 19–91%, and  $\delta$  falls across transformer depth. Shuffling the embedding-to-label correspondence leaves it unchanged, as a null-invariant quantity should. Random-weight networks give higher raw  $\delta$  but near-Gaussian features, so they are not counted as evidence; excess-based versions of these interventions await stored point clouds. On matched architecture (ViT-B supervised, DINO and CLIP) all three objectives yield a genuine excess on most datasets, and the family separations emerge with scale rather than at fixed size. High mean Ollivier–Ricci curvature coexists with tree-likeness because they live on different edges: negative curvature concentrates on between-superclass bridges, about twice as often as within superclasses on ImageNet, while the mean measures within-cluster compactness (Table 7).

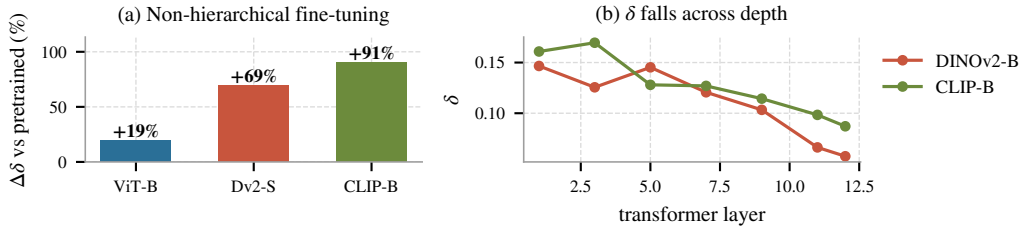


Figure 6: **The raw reading is learned: training interventions move it.** Plotted: raw  $\delta$  within architecture, (a) after non-hierarchical fine-tuning against the pretrained value, and (b) across transformer depth (layers  $\geq 1$ ).

(a) Interventions on the raw reading, within architecture.

| intervention                                     | model    | $\delta$ before | $\delta$ after | change |
|--------------------------------------------------|----------|-----------------|----------------|--------|
| non-hierarchical fine-tuning                     | DINOv2-S | 0.095           | 0.161          | +69%   |
| non-hierarchical fine-tuning                     | ViT-B    | 0.097           | 0.116          | +19%   |
| non-hierarchical fine-tuning                     | CLIP-B   | 0.104           | 0.199          | +91%   |
| across depth (layer 1 $\rightarrow$ 12)          | DINOv2-B | 0.147           | 0.058          | -61%   |
| across depth (layer 1 $\rightarrow$ 12)          | CLIP-B   | 0.161           | 0.087          | -46%   |
| random weights (pretrained $\rightarrow$ random) | DINOv2-S | 0.095           | 0.222          | +134%  |
| random weights (pretrained $\rightarrow$ random) | DINOv2-B | 0.081           | 0.198          | +144%  |
| random weights (pretrained $\rightarrow$ random) | ViT-B    | 0.113           | 0.187          | +65%   |

(b) Matched architecture, three objectives: excess of ViT-B/16 supervised, DINO and CLIP.

| dataset   | ViT-B (supervised) | DINO-B | CLIP-B |
|-----------|--------------------|--------|--------|
| ImageNet  | -0.027             | -0.004 | -0.030 |
| CIFAR-100 | -0.019             | -0.072 | -0.050 |
| CIFAR-10  | -0.081             | -0.090 | -0.069 |
| DTD       | -0.064             | -0.053 | -0.065 |
| FMNIST    | -0.071             | -0.037 | -0.051 |
| MNIST     | -0.044             | -0.069 | -0.036 |

(c) Ollivier-Ricci curvature by edge type on the centroid kNN graph (30 WordNet superclasses).

| model    | dataset | ORC within | ORC across | %neg within | %neg across |
|----------|---------|------------|------------|-------------|-------------|
| ViT-T    | IN      | 0.35       | 0.27       | 5.2         | 9.4         |
| ViT-T    | C100    | 0.51       | 0.42       | 1.4         | 2.7         |
| ViT-S    | IN      | 0.37       | 0.28       | 3.4         | 8.3         |
| ViT-S    | C100    | 0.61       | 0.48       | 0.0         | 3.0         |
| ViT-B    | IN      | 0.40       | 0.30       | 2.1         | 5.9         |
| ViT-B    | C100    | 0.61       | 0.48       | 0.0         | 1.7         |
| ViT-L    | IN      | 0.40       | 0.31       | 2.1         | 6.8         |
| ViT-L    | C100    | 0.61       | 0.47       | 0.6         | 2.1         |
| DINO-B   | IN      | 0.40       | 0.33       | 2.8         | 4.1         |
| DINO-B   | C100    | 0.60       | 0.45       | 0.0         | 1.4         |
| DINOv2-S | IN      | 0.37       | 0.33       | 1.8         | 1.2         |
| DINOv2-S | C100    | 0.60       | 0.44       | 0.0         | 0.8         |
| DINOv2-B | IN      | 0.58       | 0.64       | 0.3         | 0.1         |
| DINOv2-B | C100    | 0.66       | 0.54       | 0.0         | 0.1         |
| DINOv2-L | IN      | 0.69       | 0.72       | 0.0         | 0.0         |
| DINOv2-L | C100    | 0.70       | 0.64       | 0.0         | 0.1         |
| DINOv2-G | IN      | 0.70       | 0.74       | 0.0         | 0.0         |
| DINOv2-G | C100    | 0.77       | 0.70       | 0.0         | 0.0         |
| CLIP-B   | IN      | 0.36       | 0.29       | 4.2         | 7.9         |
| CLIP-B   | C100    | 0.60       | 0.46       | 0.0         | 3.1         |
| CLIP-L   | IN      | 0.35       | 0.27       | 4.0         | 8.4         |
| CLIP-L   | C100    | 0.57       | 0.43       | 0.0         | 2.2         |
| SigLIP-B | IN      | 0.34       | 0.26       | 5.0         | 7.5         |
| SigLIP-B | C100    | 0.59       | 0.43       | 0.0         | 2.3         |

Table 7: **The raw reading is learned, and negative curvature lives on the bridges between superclasses.** (a) Fine-tuning on a task with no class hierarchy raises  $\delta$ ,  $\delta$  falls across transformer depth, and random-weight networks read a higher raw  $\delta$  with near-Gaussian features, so they are not counted as evidence; label shuffling leaves  $\delta$  unchanged, as a null-invariant quantity should (Figure 6). (b) The three objectives on one architecture; the family separations emerge with scale rather than at fixed size. (c) Negative curvature concentrates on between-cluster bridges while the mean ORC measures within-cluster compactness; DINOv2 on ImageNet inverts the pattern and recovers it on CIFAR-100.

## B.7 IS THE STRUCTURE DEEPER THAN A STAR? THE DEPTH TEST

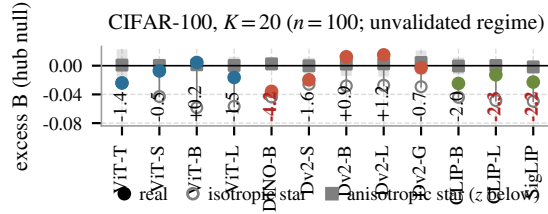


Figure 7: **The depth test on CIFAR-100 ( $K=20$ , five points per cluster) reads in the unvalidated regime.** Plotted: excess B under the hub-randomizing null for the real centroids (filled) and their matched stars (hollow circle: isotropic, withdrawn; square: anisotropic), with the  $z$  of real minus anisotropic star under each backbone.

**Real-data readings by regime.** ImageNet ( $n=1000$ , WordNet  $K=30$ ) lies in the validated regime: the real cloud is more hierarchical than its anisotropic matched star beyond the noise ( $z \leq -2$ ) in 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L), never less, under Gaussian and under Haar-resampled star hubs alike (Table 8, Figure 4a). CIFAR-100 ( $n=100$ ,  $K=20$ ) lies in the unvalidated regime, where pure stars are declared more hierarchical in 17% of runs; its readings (3 of 12 with  $z \leq -2$ , none with  $z \geq +2$ ) are reported without certification (Figure 7). Validity domain of the test as run:  $n \approx 1000$ ; at  $n=100$  the false-alarm rate is 0% with  $K=6$  (17 points per cluster) and 17% with  $K \geq 12$ , so the test is not validated below about ten points per cluster. The bar was fixed before the sweep; the restriction to the validated regime was adopted after it.

**A trained control, inconclusive.** We fine-tuned the census ViT-B/16 for two passes over the images behind its ImageNet centroids with two objectives on the same batches: cross-entropy alone, and cross-entropy plus a hierarchical cross-entropy over the WordNet superclasses. Both fine-tuned models keep the census excess of the frozen backbone ( $-0.015$  and  $-0.016$  against  $-0.016$ ) and both are certified by the depth test, with  $z$  of  $-2.33$  and  $-2.77$  against  $-3.62$  for the frozen backbone: the hierarchical term lowers  $z$  relative to plain cross-entropy but neither reaches the frozen reading, so the control neither confirms nor refutes the test (Table 8). The full training set could not be read at training speed from the archive, so the schedule is two passes over the census subset.

(a) The twelve backbones: depth = excess B of the real centroids minus that of a matched star (negative = more hierarchical above the frame), with  $z$  against the combined spread.

| model    | C100, $K=20$ , Gaussian hubs |               | IN, $K=30$ , Gaussian hubs |               | IN, $K=30$ , Haar-resampled hubs |                 |                   | bal. frame |
|----------|------------------------------|---------------|----------------------------|---------------|----------------------------------|-----------------|-------------------|------------|
|          | iso. star                    | aniso. star   | iso. star                  | aniso. star   | star B                           | depth ( $z$ )   | $r_{\text{star}}$ | real $z$   |
| ViT-T    | +0.000 (+0.0)                | -0.025 (-1.4) | -0.015 (-1.5)              | -0.018 (-1.5) | +0.0011                          | -0.0184 (-1.45) | 0/10              | -2.1       |
| ViT-S    | +0.036 (+2.3)                | -0.008 (-0.5) | -0.021 (-1.9)              | -0.028 (-2.4) | +0.0010                          | -0.0257 (-2.04) | 0/10              | -1.7       |
| ViT-B    | +0.063 (+4.5)                | +0.003 (+0.2) | -0.026 (-2.9)              | -0.033 (-3.6) | +0.0012                          | -0.0347 (-3.01) | 0/10              | -2.8       |
| ViT-L    | +0.040 (+4.2)                | -0.017 (-1.5) | -0.024 (-3.2)              | -0.033 (-4.0) | +0.0000                          | -0.0337 (-3.73) | 0/10              | -5.6       |
| DINO-B   | +0.008 (+0.9)                | -0.038 (-4.2) | -0.002 (-0.2)              | -0.009 (-0.9) | +0.0002                          | -0.0092 (-1.02) | 0/10              | -1.0       |
| DINOv2-S | +0.006 (+0.6)                | -0.020 (-1.6) | +0.011 (+0.9)              | +0.008 (+0.7) | +0.0007                          | +0.0083 (+0.71) | 10/10             | +0.1       |
| DINOv2-B | +0.041 (+5.4)                | +0.010 (+0.9) | -0.011 (-0.9)              | -0.016 (-1.2) | +0.0018                          | -0.0155 (-1.18) | 0/10              | -0.3       |
| DINOv2-L | +0.042 (+6.8)                | +0.013 (+1.2) | -0.024 (-2.3)              | -0.028 (-2.4) | -0.0007                          | -0.0267 (-2.21) | 0/10              | -1.7       |
| DINOv2-G | +0.026 (+3.8)                | -0.007 (-0.7) | -0.024 (-1.6)              | -0.029 (-1.8) | -0.0021                          | -0.0263 (-1.50) | 0/10              | -0.9       |
| CLIP-B   | +0.019 (+1.8)                | -0.024 (-2.0) | -0.006 (-0.7)              | -0.016 (-2.0) | +0.0012                          | -0.0171 (-1.48) | 1/10              | -1.3       |
| CLIP-L   | +0.036 (+5.7)                | -0.013 (-2.3) | +0.004 (+0.5)              | -0.008 (-0.8) | +0.0018                          | -0.0090 (-0.74) | 2/10              | -0.7       |
| SigLIP-B | +0.027 (+2.8)                | -0.021 (-2.2) | +0.001 (+0.0)              | -0.011 (-0.9) | +0.0024                          | -0.0121 (-0.78) | 1/10              | -1.1       |

(b) Two ViT-B/16 supervised on ImageNet-1k leaf labels only, through the census protocol, against the IN-21k census backbone.

| model                          | IN excess ( $r$ ) | C100 excess ( $r$ ) | depth, Gaussian hubs ( $z$ ) | depth, Haar hubs ( $z$ ) |
|--------------------------------|-------------------|---------------------|------------------------------|--------------------------|
| DeiT-B, IN-1k                  | -0.0073 (200)     | -0.0257 (200)       | -0.0213 (-1.26)              | -0.0197 (-1.21)          |
| ViT-B, augreg, IN-1k           | -0.0070 (199)     | -0.0117 (187)       | -0.0249 (-3.03)              | -0.0290 (-2.65)          |
| ViT-B, augreg, IN-21k (census) | -0.0162 (200)     | +0.0057 (43)        | -0.0335 (-3.62)              | -0.0347 (-3.01)          |

Table 8: **Hierarchy above the superclasses is certified in four ImageNet backbones under both matched stars, imposing the geometry does not create it, and leaf-label supervision can produce it or not.** (a) Isotropic star: Gaussian clouds with each superclass’s RMS spread (the earlier, withdrawn construction); anisotropic star: within each superclass a Haar sample with the cloud’s own covariance, hubs Gaussian at the real hub radius; Haar-resampled hubs: the same clusters with the hubs drawn as a Haar resample of the real hubs. Hub-randomizing Haar null, 10 star seeds; frames: CIFAR-100 coarse labels, ImageNet WordNet cut. The test is validated at  $n=1000$  only (Table 9); CIFAR-100 readings ( $n=100$ ) are reported without certification. Certified on ImageNet ( $n=1000$ , the validated regime): 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L) with  $z$  from  $-2.4$  to  $-4.0$ ; the withdrawn isotropic star reached  $z=+6.8$  on CIFAR-100. With the hubs of the matched star drawn as a Haar resample of the real hubs (exact hub spectrum, 10 star seeds) the certified set is ViT-S, ViT-B, ViT-L, DINOv2-L, with  $z$  from  $-2.04$  to  $-3.73$ ;  $r_{\text{star}}$  counts the star seeds whose excess B lies at or below the real one, a rank whose resolution is one part in eleven and which does not separate the certified backbones from the others. Balanced frame (complete-linkage WordNet cut with the smallest variance of cluster sizes, chosen before any depth run): certified ViT-T, ViT-B, ViT-L. K sweep: CIFAR-100  $K=5$ : iso 0/12 at  $z \leq -2$ , 7/12 at  $z \geq +2$ ; aniso 0/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . CIFAR-100  $K=10$ : iso 0/12 at  $z \leq -2$ , 9/12 at  $z \geq +2$ ; aniso 1/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . CIFAR-100  $K=20$ : iso 0/12 at  $z \leq -2$ , 8/12 at  $z \geq +2$ ; aniso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=10$ : iso 0/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 2/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=30$ : iso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 4/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . ImageNet  $K=60$ : iso 3/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ ; aniso 4/12 at  $z \leq -2$ , 0/12 at  $z \geq +2$ . (b) Supervision on leaf labels can produce the depth (the augreg recipe is certified under both stars) and need not (DeiT-B is not detected): the hierarchical label set of IN-21k is not what makes the census ViTs deep; their WordNet alignment is in Table 11.

(c) A hyperbolic backbone and its Euclidean twin under the census of record and the depth test.

|            |         | CIFAR-100 ( $K=20$ , $n=100$ : depth unvalidated) |                                  |           | ImageNet ( $K=30$ , $n=1000$ : depth validated) |                                  |           |
|------------|---------|---------------------------------------------------|----------------------------------|-----------|-------------------------------------------------|----------------------------------|-----------|
| model      | input   | exc. ( $r$ )                                      | $\hat{\delta}_{99.9}$ nat./Eucl. | depth $z$ | exc. ( $r$ )                                    | $\hat{\delta}_{99.9}$ nat./Eucl. | depth $z$ |
| MERU ViT-S | images  | −0.020 (199)                                      | 0.117/0.117                      | −1.5      | −0.008° (186)                                   | 0.088/0.088                      | −1.5      |
| CLIP ViT-S | images  | −0.028 (200)                                      | 0.106/0.107                      | −3.9      | −0.006° (171)                                   | 0.110/0.088                      | −1.8      |
| MERU ViT-B | images  | −0.020 (200)                                      | 0.114/0.114                      | −2.2      | −0.010 (196)                                    | 0.087/0.087                      | −0.9      |
| CLIP ViT-B | images  | −0.030 (200)                                      | 0.105/0.103                      | −3.3      | −0.009° (187)                                   | 0.107/0.085                      | −1.0      |
| MERU ViT-L | images  | −0.012° (195)                                     | 0.118/0.118                      | −0.3      | −0.007° (184)                                   | 0.088/0.088                      | −0.5      |
| CLIP ViT-L | images  | −0.016 (199)                                      | 0.123/0.112                      | −1.7      | −0.010 (197)                                    | 0.104/0.084                      | −1.2      |
| MERU ViT-S | prompts | −0.022 (200)                                      | 0.086/0.086                      | −0.7      | −0.003° (155)                                   | 0.085/0.086                      | +0.0      |
| CLIP ViT-S | prompts | −0.023 (200)                                      | 0.085/0.082                      | −0.8      | +0.006° (15)                                    | 0.093/0.087                      | +0.8      |
| MERU ViT-B | prompts | −0.012 (200)                                      | 0.096/0.096                      | −0.7      | −0.001° (132)                                   | 0.089/0.090                      | +0.8      |
| CLIP ViT-B | prompts | −0.017 (200)                                      | 0.094/0.088                      | +0.3      | +0.008° (2)                                     | 0.100/0.092                      | +0.2      |
| MERU ViT-L | prompts | −0.006° (190)                                     | 0.102/0.102                      | +0.6      | +0.001° (75)                                    | 0.093/0.093                      | +0.5      |
| CLIP ViT-L | prompts | −0.012 (200)                                      | 0.098/0.092                      | +0.5      | +0.006° (5)                                     | 0.101/0.092                      | +0.4      |

(d) A trained control: the census ViT-B/16 fine-tuned with and without a hierarchical objective.

| ViT-B/16                                     | $\hat{\delta}_{99.9}$ | excess  | $r/200$ | depth   | $z$   |
|----------------------------------------------|-----------------------|---------|---------|---------|-------|
| frozen (census)                              | 0.054                 | −0.0162 | 200     | −0.0335 | −3.62 |
| cross-entropy                                | 0.048                 | −0.0153 | 200     | −0.0303 | −2.33 |
| cross-entropy + hierarchical CE (WordNet-30) | 0.048                 | −0.0162 | 200     | −0.0295 | −2.77 |

Table 8: (continued) (c) MERU (Desai et al., 2023) embeds images and text on the Lorentz hyperboloid with an entailment objective; its released twin is a CLIP baseline with the same ViT, RedCaps data and recipe minus the hyperbolic lift and entailment loss. Both pass through the instrument unchanged: class centroids of the projected embeddings (MERU: space-like hyperboloid coordinates; CLIP: unit vectors), Euclidean distances; exc. ( $r$ ): excess over the Haar spectrum-matched null with the number of 200 replicates above the real value (°: not genuine under BH over the 24 cells); nat./Eucl.: the same statistic in the model’s own metric (Lorentz distance between tangent-space-mean centroids for MERU, angular for CLIP) against the Euclidean one; depth  $z$ : the anisotropic matched-star test of panel (a) (10 star seeds), validated at  $n=1000$  only. On ImageNet images the excess runs from −0.007 to −0.010 for MERU and from −0.006 to −0.010 for its twin across ViT-S/B/L; MERU’s Lorentz-metric  $\hat{\delta}_{99.9}$  differs from its Euclidean value by at most 0.0002; depth  $z$  runs from −1.5 to −0.5 for MERU and from −1.8 to −1.0 for CLIP, so neither is more hierarchical than a matched star and MERU is never ahead of its twin. Its embeddings sit in the near-flat regime of the hyperboloid: learned curvature  $c=0.10$ , spatial norm times  $\sqrt{c}$  at the median 0.258–0.279 (95th percentile 0.263–0.287), and Lorentz-to-Euclidean pairwise distance ratio 0.9972–0.9991 at the median (0.9988–0.9997 on the class centroids), so the control tests the training objective, not a curved geometry. MERU’s hierarchy is a generic→specific (text⊃image) partial order, not a class taxonomy. (d) Two passes over the 100 training images per class behind the ImageNet centroids (patch embedding frozen, AdamW, same batches and seed for both objectives), with plain cross-entropy or with an added hierarchical cross-entropy over the WordNet 30-cut; both fine-tuned models stay certified and neither reaches the frozen reading, so the control is inconclusive.

## B.8 HOW MUCH DEPTH CAN THE TEST SEE? POWER ON SYNTHETIC AND IMPLANTED HIERARCHIES

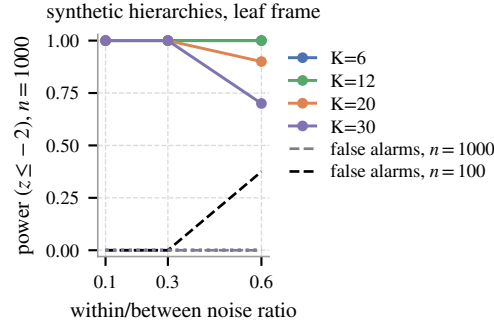


Figure 8: **The depth test has power on synthetic hierarchies at ImageNet’s class count when the within-cluster noise is below the between-hub spread.** Plotted: power ( $z \leq -2$ ) on synthetic two- and three-level hierarchies with the frame at the leaf clusters (solid, one line per  $K$ ) or at the top level (dotted), against the within/between noise ratio, with the false-alarm rates of pure stars dashed.

(a) Synthetic hierarchies at  $d=768$ : two- and three-level hierarchies (pooled) and pure stars, within/between noise ratios 0.1/0.3/0.6, isotropic and anisotropic clusters pooled, 5 seeds.

| frame                           | $n$  | $K$ | power @ ratio 0.1 | 0.3  | 0.6  | star $z \leq -2$ | star $z \geq +2$ |
|---------------------------------|------|-----|-------------------|------|------|------------------|------------------|
| top-level, isotropic star       | 100  | 6   | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 100  | 12  | 0.00              | 0.00 | 0.00 | 0.00             | 0.13             |
| top-level, isotropic star       | 100  | 20  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 100  | 30  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 6   | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 12  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 20  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| top-level, isotropic star       | 1000 | 30  | 0.00              | 0.00 | 0.00 | 0.00             | 0.17             |
| leaf clusters, anisotropic star | 100  | 6   | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 100  | 12  | 1.00              | 1.00 | 1.00 | 0.17             | 0.00             |
| leaf clusters, anisotropic star | 100  | 20  | 1.00              | 1.00 | 1.00 | 0.17             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 6   | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 12  | 1.00              | 1.00 | 1.00 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 20  | 1.00              | 1.00 | 0.90 | 0.00             | 0.00             |
| leaf clusters, anisotropic star | 1000 | 30  | 1.00              | 1.00 | 0.70 | 0.00             | 0.00             |

(b) Implanted two-level trees on the real ImageNet centroids: the hub arrangement replaced at strength  $s$ , every within-cluster offset kept; hits = implant seeds with  $z \leq -2$  at  $s=0/0.5/1$ ,  $\bar{z}_1$  = mean  $z$  at  $s=1$ , w/b = within/between spread.

| model    | WordNet-30 frame of record, Gaussian-hub star |     |       |       |             |                          | Haar-hub star |             | balanced frame (pre-specified) |       |       |
|----------|-----------------------------------------------|-----|-------|-------|-------------|--------------------------|---------------|-------------|--------------------------------|-------|-------|
|          | real $z$                                      | w/b | hits  | $s^*$ | $\bar{z}_1$ | tight $z$ at $s=0/0.5/1$ | hits $s=0/1$  | $\bar{z}_1$ | real $z$                       | hits  | $s^*$ |
| ViT-T    | -1.5                                          | 1.5 | 0/0/0 | -     | +0.1        | -0.6/-1.9/-9.2           | 0/0           | +0.0        | -2.1                           | 0/0/1 | -     |
| ViT-S    | -2.4                                          | 1.9 | 0/0/0 | -     | -1.2        | -0.2/-3.4/-10.1          | 0/0           | -0.7        | -1.7                           | 0/0/0 | -     |
| ViT-B    | -3.6                                          | 2.0 | 0/0/0 | -     | +0.4        | -1.0/-3.9/-11.5          | 0/0           | +0.4        | -2.8                           | 0/0/0 | -     |
| ViT-L    | -4.0                                          | 1.9 | 0/0/0 | -     | -1.4        | -1.6/-4.4/-12.7          | 0/0           | -1.0        | -5.6                           | 0/0/0 | -     |
| DINO-B   | -0.9                                          | 2.1 | 0/0/0 | -     | -0.3        | -1.6/-5.1/-12.1          | 0/0           | -0.3        | -1.0                           | 0/0/0 | -     |
| DINOv2-S | +0.7                                          | 3.0 | 0/0/0 | -     | -0.8        | -0.2/-3.1/-11.5          | 0/0           | -0.8        | +0.1                           | 0/0/0 | -     |
| DINOv2-B | -1.2                                          | 3.7 | 0/0/0 | -     | +0.4        | -0.0/-3.1/-11.2          | 0/0           | +0.5        | -0.3                           | 0/0/0 | -     |
| DINOv2-L | -2.4                                          | 3.9 | 0/0/0 | -     | +0.2        | +0.2/-2.7/-11.4          | 0/0           | +0.2        | -1.7                           | 0/0/0 | -     |
| DINOv2-G | -1.8                                          | 3.9 | 0/0/0 | -     | +1.3        | +0.2/-3.9/-11.6          | 0/0           | +1.2        | -0.9                           | 0/0/0 | -     |
| CLIP-B   | -2.0                                          | 1.3 | 0/0/0 | -     | -1.2        | -1.7/-3.8/-11.4          | 0/0           | -0.7        | -1.3                           | 0/0/0 | -     |
| CLIP-L   | -0.8                                          | 1.5 | 0/0/3 | -     | -2.2        | -2.4/-5.8/-13.5          | 0/1           | -1.7        | -0.7                           | 0/0/0 | -     |
| SigLIP-B | -0.9                                          | 1.5 | 0/0/0 | -     | -1.6        | -2.1/-5.0/-11.0          | 0/0           | -1.1        | -1.1                           | 0/0/0 | -     |

Table 9: **The depth test meets its bar on synthetic hierarchies at ImageNet’s class count, raises no false alarm on real clouds under either star, and misses implanted depth at the real within-cluster spread.** (a) Power = fraction of hierarchy runs with  $z \leq -2$ ; star columns = false-alarm rates in each direction. Top-level frame: the test receives the  $K$  super-cluster labels, as the earlier isotropic test did on the real backbones; its hub null keeps every frame cluster intact and only rearranges the  $K$  hubs, so hierarchy below the frame is invisible by construction. Leaf frame: the test receives the finest cluster labels with the anisotropic matched star (configurations with more leaves than points are infeasible and omitted). Pre-set bar: power  $\geq 0.8$  for two- and three-level hierarchies at noise ratios  $\leq 0.3$  for both  $n$ , and false alarms  $\leq 5\%$  in each direction. Leaf frame: met at  $n=1000$  (power 1.00 at ratios  $\leq 0.3$  and 0.90 at 0.6; false alarms 0% and 0%), failed at  $n=100$ , where  $K \geq 12$  (under ten points per cluster) gives 17% false alarms at  $z \leq -2$ . Top-level frame: power 0.00, false alarms 16% pooled. (b) Six super-hubs of five, hub =  $s$ -super-hub +  $(1 - s + 0.4s)$ -own Gaussian draw, hub cloud rescaled to the real RMS radius; the depth test of Table 8 runs unchanged, 5 implant seeds per  $s$ . hits: implant seeds with  $z \leq -2$  (of 5);  $s^*$ : first  $s$  with  $\geq 4$  of 5; within/between: RMS of the within-cluster offsets over the point-weighted RMS of the hub displacements (1.3 to 3.9 on the real clouds, against 0.1–0.6 in the synthetic sweep); tight: the same implant after shrinking the offsets to within/between = 0.6 (2 seeds). Power (fraction of runs with  $z \leq -2$ ) by  $s$ :  $s=0$ : 0.00,  $s=0.25$ : 0.00,  $s=0.5$ : 0.00,  $s=0.75$ : 0.00,  $s=1$ : 0.05; false alarms at  $s=0$ : 0 of 60. Under the Haar-hub star the same implants give 0 of 60 false alarms at  $s=0$  and 1 of 60 detections at  $s=1$  (power 0.02). Balanced frame: the  $K=30$  agglomerative cut of the WordNet distance matrix (complete linkage) with the smallest variance of cluster sizes, chosen and logged before any depth run; sizes 1–154 classes against 1–316 for the frame of record.

### B.9 IS THE ISLAND REAL? THE TREE MAP UNDER SIX CONFIGURATIONS

Threshold sensitivity of the selection criterion of §5.4: the selected configuration is stable for degeneracy bounds in  $[0.45, 0.50]$  on ImageNet and  $[0.40, 0.50]$  on CIFAR-100; tighter bounds select cosine-complete on ImageNet (gap 0.13) and cosine-Ward on CIFAR-100 (gap 0.56), and in every case the label-free gap is moderate and configuration-sensitive, never a categorical island. The criterion must be applied per dataset: the configuration it selects on DBpedia (Euclidean, average linkage, where nothing is degenerate) is exactly the degenerate naive configuration on ImageNet, so a criterion transferred across datasets would reintroduce the island.

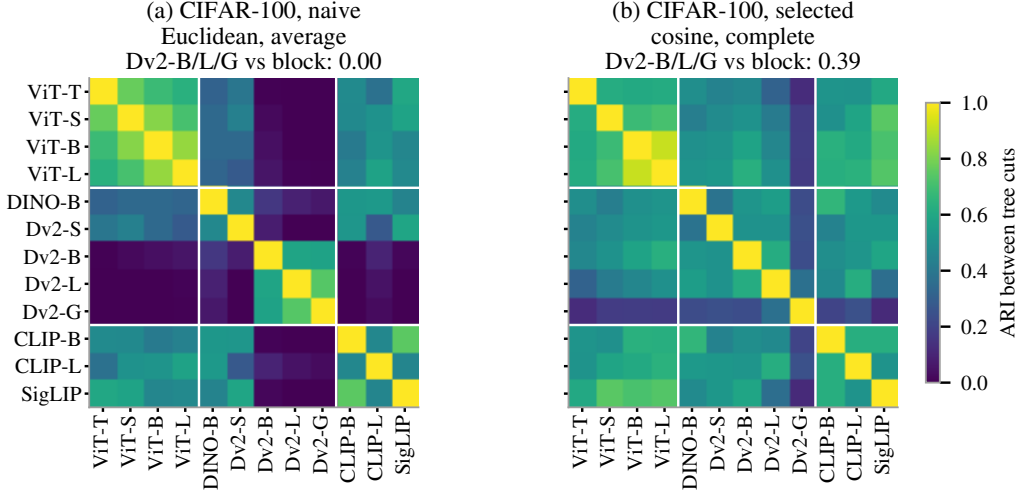


Figure 9: **The island is an artifact on CIFAR-100 too.** Plotted: the same reading as Figure 5 on the 20 CIFAR-100 superclass cuts, naive (a) against criterion-selected (b).

(a) Selection diagnostics and mean pairwise ARI at the reference cut.

| dataset   | configuration              | max frac. | CPCC  | mean pairwise ARI at the reference cut |                    |        |
|-----------|----------------------------|-----------|-------|----------------------------------------|--------------------|--------|
|           |                            |           |       | B/L/G vs block                         | S, DINO-B vs block | within |
| ImageNet  | euclid-average (deg.)      | 0.97      | 0.761 | 0.03                                   | 0.08               | 0.64   |
| ImageNet  | euclid-complete (deg.)     | 0.83      | 0.606 | 0.03                                   | 0.17               | 0.44   |
| ImageNet  | euclid-ward                | 0.48      | 0.464 | 0.15                                   | 0.38               | 0.53   |
| ImageNet  | cosine-average (selected)  | 0.40      | 0.691 | 0.38                                   | 0.35               | 0.48   |
| ImageNet  | cosine-complete            | 0.32      | 0.539 | 0.13                                   | 0.17               | 0.50   |
| ImageNet  | cosine-ward                | 0.37      | 0.464 | 0.17                                   | 0.39               | 0.55   |
| CIFAR-100 | euclid-average (deg.)      | 0.80      | 0.798 | 0.00                                   | 0.41               | 0.57   |
| CIFAR-100 | euclid-complete (deg.)     | 0.66      | 0.722 | 0.17                                   | 0.40               | 0.54   |
| CIFAR-100 | euclid-ward                | 0.33      | 0.534 | 0.37                                   | 0.65               | 0.63   |
| CIFAR-100 | cosine-average (deg.)      | 0.53      | 0.777 | 0.30                                   | 0.54               | 0.58   |
| CIFAR-100 | cosine-complete (selected) | 0.39      | 0.698 | 0.39                                   | 0.51               | 0.63   |
| CIFAR-100 | cosine-ward                | 0.14      | 0.584 | 0.56                                   | 0.62               | 0.61   |

(b) Three measures of DINOv2-B/L/G against the supervised+contrastive block, each written as vs block / within block.

| dataset   | configuration              | ARI at the cut | cophenetic corr. | triplet agreement |
|-----------|----------------------------|----------------|------------------|-------------------|
| ImageNet  | euclid-average (deg.)      | 0.03 / 0.64    | 0.36 / 0.80      | 0.47 / 0.74       |
| ImageNet  | euclid-complete (deg.)     | 0.03 / 0.44    | 0.20 / 0.66      | 0.41 / 0.65       |
| ImageNet  | euclid-ward                | 0.16 / 0.53    | 0.62 / 0.84      | 0.67 / 0.85       |
| ImageNet  | cosine-average (selected)  | 0.39 / 0.48    | 0.48 / 0.78      | 0.77 / 0.76       |
| ImageNet  | cosine-complete            | 0.13 / 0.49    | 0.18 / 0.72      | 0.36 / 0.71       |
| ImageNet  | cosine-ward                | 0.17 / 0.55    | 0.59 / 0.85      | 0.69 / 0.86       |
| CIFAR-100 | euclid-average (deg.)      | 0.00 / 0.57    | 0.15 / 0.78      | 0.41 / 0.74       |
| CIFAR-100 | euclid-complete (deg.)     | 0.17 / 0.54    | 0.23 / 0.72      | 0.41 / 0.66       |
| CIFAR-100 | euclid-ward                | 0.37 / 0.63    | 0.63 / 0.77      | 0.70 / 0.79       |
| CIFAR-100 | cosine-average (deg.)      | 0.30 / 0.58    | 0.42 / 0.72      | 0.51 / 0.64       |
| CIFAR-100 | cosine-complete (selected) | 0.39 / 0.63    | 0.46 / 0.64      | 0.50 / 0.61       |
| CIFAR-100 | cosine-ward                | 0.56 / 0.61    | 0.74 / 0.73      | 0.76 / 0.76       |

Table 10: **The categorical island is a chaining artifact of the naive cut; a moderate gap remains under every admissible configuration and closes only in triplet agreement under cosine-average.** (a) Six configurations (metric  $\times$  linkage) per dataset with the two quantities that drive the selection criterion of §5.4: the degeneracy diagnostic (largest-cluster fraction at the reference cut, worst model) and the mean cophenetic fidelity (CPCC) of each model’s dendrogram to its own distance matrix; configurations with max-cluster fraction  $> 0.5$  are excluded and, among the rest, the highest CPCC selects the marked row; then the mean pairwise ARI between the cuts at 30 (ImageNet) or 20 (CIFAR-100) clusters. (b) Dendrograms rebuilt from the census cache: ARI at the cut, Pearson correlation between the two trees’ cophenetic distance vectors, and agreement on which pair merges first over  $10^4$  random class triplets. Under the naive ImageNet configuration, cross-family ARI between the supervised ViTs and the contrastive VLMs reaches 0.82 while every DINOv2-vs-block pair stays at or below 0.06, and the worst model’s largest cluster holds 97% of the classes. Under the selected configuration the DINOv2 family agrees with the block at 0.38 against 0.48 within the block. Under the other 3 admissible ImageNet configurations the DINOv2-vs-block agreement is 0.13–0.17 against 0.50–0.55 within the block; over the admissible configurations and the three measures the median fraction of within-block agreement missing is 0.34, about a third.

#### B.10 WHOSE TAXONOMY? WORDNET ALIGNMENT, SUPERCLASS RECOVERY, DBPEDIA AND HIERARCAPS

On HierarCaps (Alper & Averbuch-Elor, 2024) (1000 four-level caption chains), text encoders show the same reading with no class set: the projection radius orders levels within a chain, sibling triplets are resolved at 0.83–0.97 (chance 0.5), and the leaf excess repeats the text census (Table 11).

| (a) Alignment with WordNet on ImageNet.                                                                                                                                                              |                                                      |             |                      |                  |                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------|----------------------|------------------|-----------------------------------------------|
| model                                                                                                                                                                                                | Spearman $\rho$ , inter-centroid vs WordNet distance |             |                      |                  | raw $\hat{\delta}$ of WordNet / random groups |
|                                                                                                                                                                                                      | $\rho_{WN}$                                          | shuffle     | Gaussian             | single image     |                                               |
| ViT-T                                                                                                                                                                                                | +0.52                                                | +0.02       | +0.01                | –                | 0.146 / 0.159                                 |
| ViT-S                                                                                                                                                                                                | +0.50                                                | +0.00       | –0.00                | –                | 0.110 / 0.130                                 |
| ViT-B                                                                                                                                                                                                | +0.49                                                | +0.03       | +0.01                | –                | 0.092 / 0.120                                 |
| ViT-L                                                                                                                                                                                                | +0.53                                                | +0.01       | +0.03                | +0.25            | 0.104 / 0.117                                 |
| DINO-B                                                                                                                                                                                               | +0.36                                                | –0.01       | –0.01                | –                | 0.084 / 0.117                                 |
| DINOv2-S                                                                                                                                                                                             | +0.22                                                | +0.01       | +0.01                | –                | 0.077 / 0.100                                 |
| DINOv2-B                                                                                                                                                                                             | +0.18                                                | –0.01       | +0.01                | –                | 0.068 / 0.069                                 |
| DINOv2-L                                                                                                                                                                                             | +0.18                                                | –0.00       | –0.03                | +0.08            | 0.060 / 0.058                                 |
| DINOv2-G                                                                                                                                                                                             | +0.20                                                | –0.03       | –0.01                | +0.03            | 0.054 / 0.056                                 |
| CLIP-B                                                                                                                                                                                               | +0.57                                                | +0.00       | +0.01                | –                | 0.126 / 0.174                                 |
| CLIP-L                                                                                                                                                                                               | +0.59                                                | +0.01       | +0.03                | +0.26            | 0.117 / 0.162                                 |
| SigLIP-B                                                                                                                                                                                             | +0.58                                                | –0.01       | +0.01                | –                | 0.118 / 0.158                                 |
| DeiT-B (IN-1k)                                                                                                                                                                                       | +0.08                                                | +0.01       | –                    | –                | –                                             |
| ViT-B (augreg, IN-1k)                                                                                                                                                                                | +0.52                                                | +0.01       | –                    | –                | –                                             |
| (b) Recovery of the CIFAR-100 superclasses: ARI of the 20-cluster cut per configuration ( $\dagger$ : degenerate on CIFAR-100, euclid-average being the naive configuration; *: criterion-selected). |                                                      |             |                      |                  |                                               |
| model                                                                                                                                                                                                | euclid-avg $\dagger$                                 | euclid-ward | cosine-avg $\dagger$ | cosine-complete* | cosine-ward                                   |
| ViT-T                                                                                                                                                                                                | 0.31                                                 | 0.38        | 0.29                 | 0.33             | 0.33                                          |
| ViT-S                                                                                                                                                                                                | 0.35                                                 | 0.56        | 0.37                 | 0.47             | 0.56                                          |
| ViT-B                                                                                                                                                                                                | 0.39                                                 | 0.61        | 0.43                 | 0.45             | 0.57                                          |
| ViT-L                                                                                                                                                                                                | 0.38                                                 | 0.55        | 0.49                 | 0.48             | 0.60                                          |
| DINO-B                                                                                                                                                                                               | 0.22                                                 | 0.48        | 0.28                 | 0.38             | 0.54                                          |
| DINOv2-S                                                                                                                                                                                             | 0.12                                                 | 0.51        | 0.32                 | 0.33             | 0.49                                          |
| DINOv2-B                                                                                                                                                                                             | 0.02                                                 | 0.47        | 0.35                 | 0.42             | 0.51                                          |
| DINOv2-L                                                                                                                                                                                             | 0.01                                                 | 0.32        | 0.25                 | 0.50             | 0.48                                          |
| DINOv2-G                                                                                                                                                                                             | 0.00                                                 | 0.19        | 0.08                 | 0.20             | 0.52                                          |
| CLIP-B                                                                                                                                                                                               | 0.18                                                 | 0.48        | 0.18                 | 0.37             | 0.48                                          |
| CLIP-L                                                                                                                                                                                               | 0.35                                                 | 0.52        | 0.32                 | 0.41             | 0.54                                          |
| SigLIP-B                                                                                                                                                                                             | 0.22                                                 | 0.49        | 0.22                 | 0.37             | 0.52                                          |
| DeiT-B (IN-1k)                                                                                                                                                                                       | 0.22                                                 | 0.56        | 0.23                 | 0.38             | 0.54                                          |
| ViT-B (augreg, IN-1k)                                                                                                                                                                                | 0.30                                                 | 0.44        | 0.40                 | 0.42             | 0.48                                          |

Table 11: **Alignment with the human taxonomy follows supervision and recipe, the superclasses are recovered by every family, and the recovery is not WordNet circularity.** (a)  $\rho_{WN}$ : Spearman correlation between inter-centroid and WordNet tree distances on ImageNet, with the label-shuffle and Gaussian-cloud controls; single image: the same correlation on single-image distances (one image per class); WordNet / random groups: raw  $\hat{\delta}$  of WordNet-coherent against random class groupings. (b) ARI of the 20-cluster cut against the true CIFAR-100 superclasses per configuration (degenerate configurations marked; the criterion-selected one is cosine-complete). Paired within each of the 3 admissible configurations, the label-supervised ViT-B recovers more than each DINOv2 model in 11 of 12 comparisons; DINOv2 itself recovers the superclasses at ARI 0.19–0.52, and the best cut over models and configurations reaches 0.61 against zero for permuted labels. Circularity controls: WordNet-coherent groupings read a lower raw  $\delta$  than random ones in 11 of 12 models, and single-image distances still correlate with WordNet at up to +0.26. The two leaf-label ViT-B/16 of Table 8: the augreg recipe is aligned with WordNet like the IN-21k ViTs ( $\rho_{WN}$  +0.52) while DeiT-B is nearly unaligned (+0.08) yet recovers the CIFAR-100 superclasses at ARI up to 0.56, so WordNet alignment is recipe-dependent even among leaf-supervised ViTs.

(c) DBpedia Classes (219 leaf classes, 3 levels), a text hierarchy independent of WordNet; gains in pp.

| model    | supremum, Gaussian    |        | census of record      |        |             | trip. | NC H-R | FS H-R     |
|----------|-----------------------|--------|-----------------------|--------|-------------|-------|--------|------------|
|          | $\hat{\delta}_{\max}$ | excess | $\hat{\delta}_{99.9}$ | excess | $r(p)$      |       |        |            |
| BGE-base | 0.122                 | -0.028 | 0.082                 | -0.021 | 200 (0.005) | 0.89  | -0.59  | +0.05±0.05 |
| E5-base  | 0.127                 | -0.024 | 0.086                 | -0.017 | 200 (0.005) | 0.89  | -0.43  | +0.04±0.03 |
| GTE-base | 0.132                 | -0.020 | 0.088                 | -0.017 | 200 (0.005) | 0.89  | -0.82  | +0.08±0.06 |

(d) HierarCaps (1000 four-level caption chains): a multi-level text hierarchy with no class set.

| model    | $\rho(\text{level}, \text{radius})$ | % monotone | trip. L1 | trip. L2 | $\hat{\delta}$ leaves | excess |
|----------|-------------------------------------|------------|----------|----------|-----------------------|--------|
| BGE-base | +0.80±0.31                          | 49.6       | 0.88     | 0.97     | 0.130                 | +0.003 |
| E5-base  | +0.87±0.19                          | 53.9       | 0.90     | 0.97     | 0.139                 | +0.010 |
| GTE-base | +0.47±0.55                          | 24.0       | 0.90     | 0.97     | 0.135                 | +0.006 |
| CLIP-B   | +0.64±0.43                          | 27.2       | 0.86     | 0.95     | 0.123                 | -0.011 |
| CLIP-L   | +0.51±0.49                          | 18.7       | 0.86     | 0.96     | 0.117                 | -0.010 |
| SigLIP-B | +0.32±0.55                          | 9.8        | 0.83     | 0.91     | 0.138                 | +0.014 |

Table 11: (continued) (c) Excess under the original supremum reading (3 Gaussian replicates) and under the census of record (Haar null, 99.9th-percentile statistic, 200 replicates;  $r$  = replicates above the real value, left-tail  $p$ ), sibling-triplet agreement, and the metric gains, marginal as the diagnostic predicts. Tree map on the same classes: no configuration degenerates (largest cluster  $\leq 10\%$ ), the embedders recover the true 70-way level-2 partition at ARI 0.21–0.29 in all 6 configurations and agree with each other at cross-model ARI 0.65–0.82; the configuration the criterion selects there (Euclidean, average linkage) is the one that degenerates on ImageNet, which is why the criterion is applied per dataset. (d) Radial ordering of the four levels under the paper’s projection (chance for strict monotonicity =  $1/24 \approx 4.2\%$ ; shuffle controls  $\approx 0$ ), sibling triplet agreement (cosine; chance 0.5), and leaf  $\hat{\delta}$  with spectrum-null excess.

#### B.11 IS TEXT THE SAME? THE TEXT CENSUS UNDER THE RECORD

**Extraction, anisotropy and templates.** Extraction is part of the measurement: left-padded batching moves GPT-2’s  $\hat{\delta}$  by up to 0.018, of the order of the fragile excesses; the padding-free protocol is the record. Genuineness is not a proxy for anisotropy: GTE-base has the highest effective rank and is not genuine, and the only LLM-backed embedder tested (GTE-Qwen2-1.5B) is not genuine under the record. Causal-LM values are template-sensitive but the scale ordering is template-robust (Table 12).

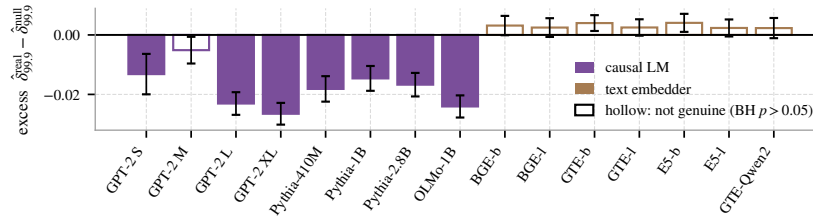


Figure 10: **In text, clustered structure depends on recipe and scale: robust at the larger GPT-2 sizes, fragile at the smaller.** Plotted: the text census of record (padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates), with error bars combining null and estimator s.d., hollow bars for cells that are not genuine, and OLMo-7B omitted as not re-extracted.

(a) The census of record on the 1000 ImageNet class prompts: padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates.

| model          | type      | $d$  | $\hat{\delta}_{99.9}$                    | excess  | record<br>exc./null | $r/200$ | $p$  | $2 \times 2$ | cosine<br>exc. | $r$ |
|----------------|-----------|------|------------------------------------------|---------|---------------------|---------|------|--------------|----------------|-----|
| GPT-2 S        | causal LM | 768  | 0.077                                    | -0.013  | -15%                | 197     | .020 | ●○○○         | +0.000°        | 101 |
| GPT-2 M        | causal LM | 1024 | 0.052                                    | -0.005° | -9%                 | 170     | .154 | ○○○○         | -0.019         | 200 |
| GPT-2 L        | causal LM | 1280 | 0.053                                    | -0.023  | -30%                | 200     | .005 | ●●●●         | -0.016         | 200 |
| GPT-2 XL       | causal LM | 1600 | 0.048                                    | -0.027  | -36%                | 200     | .005 | ●●●●         | -0.018         | 200 |
| Pythia-410M    | causal LM | 1024 | 0.067                                    | -0.018  | -21%                | 200     | .005 | ●●●●         | -0.024         | 200 |
| Pythia-1B      | causal LM | 2048 | 0.066                                    | -0.015  | -18%                | 199     | .010 | ●●●●         | -0.026         | 200 |
| Pythia-2.8B    | causal LM | 2560 | 0.066                                    | -0.017  | -20%                | 200     | .005 | ●●●●         | -0.023         | 200 |
| OLMo-1B        | causal LM | 2048 | 0.051                                    | -0.024  | -32%                | 200     | .005 | ●●●●         | -0.032         | 200 |
| OLMo-7B        | causal LM | 4096 | not re-extracted (exceeds the local GPU) |         |                     |         |      |              |                |     |
| BGE-base       | embedder  | 768  | 0.078                                    | +0.003° | +4%                 | 26      | .871 | ○○○○         | -0.002°        | 151 |
| BGE-large      | embedder  | 1024 | 0.078                                    | +0.002° | +3%                 | 49      | .756 | ○○○○         | -0.003°        | 170 |
| GTE-base       | embedder  | 768  | 0.076                                    | +0.004° | +6%                 | 11      | .945 | ○○○○         | +0.001°        | 79  |
| GTE-large      | embedder  | 1024 | 0.075                                    | +0.003° | +3%                 | 36      | .821 | ○○○○         | -0.001°        | 131 |
| GTE-Qwen2-1.5B | embedder  | 1536 | 0.078                                    | +0.002° | +3%                 | 58      | .711 | ○○○○         | -0.006         | 199 |
| E5-base        | embedder  | 768  | 0.074                                    | +0.004° | +6%                 | 15      | .925 | ○○○○         | +0.000°        | 97  |
| E5-large       | embedder  | 1024 | 0.071                                    | +0.002° | +3%                 | 44      | .781 | ○○○○         | -0.001°        | 119 |

(b) Prompt templates: raw  $\hat{\delta}$  per template and, for the four GPT-2 sizes, the template-conditioned excess over the spectrum null (3 replicates; \*:  $|z| \geq 2$ ) and its fraction of the null.

| template                  | raw $\hat{\delta}$ |       |         |         | GPT-2 excess (padding-free) |         |         |          | GPT-2 excess / null |      |      |      |
|---------------------------|--------------------|-------|---------|---------|-----------------------------|---------|---------|----------|---------------------|------|------|------|
|                           | BGE-b              | E5-b  | GPT-2 S | GPT-2 M | S                           | M       | L       | XL       | S                   | M    | L    | XL   |
| photo                     | 0.124              | 0.114 | 0.131   | 0.091   | -0.016*                     | -0.019* | -0.043* | -0.050*  | -11%                | -19% | -34% | -41% |
| name_only                 | 0.118              | 0.119 | 0.145   | 0.122   | -0.006                      | -0.007  | -0.048* | -0.051*  | -6%                 | -5%  | -39% | -43% |
| image                     | 0.125              | 0.117 | 0.110   | 0.110   | -0.018*                     | -0.028* | -0.050* | -0.049*  | -13%                | -26% | -41% | -41% |
| closeup                   | 0.126              | 0.120 | 0.086   | 0.100   | +0.007                      | -0.018* | -0.044* | -0.049*  | +5%                 | -18% | -36% | -40% |
| this_is                   | 0.125              | 0.115 | 0.080   | 0.134   | -0.032*                     | -0.022* | -0.042* | -0.046*  | -19%                | -22% | -34% | -38% |
| wild                      | 0.115              | 0.116 | 0.089   | 0.104   | -0.023*                     | -0.067* | -0.014  | -0.010*  | -16%                | -43% | -11% | -8%  |
| def_pair                  | 0.119              | 0.108 | 0.105   | 0.081   | -0.002                      | -0.019* | -0.036* | -0.035*  | -1%                 | -21% | -28% | -29% |
| gloss_only                | 0.115              | 0.115 | 0.124   | 0.107   | -0.020*                     | -0.030* | -0.033* | -0.036*  | -14%                | -31% | -27% | -29% |
| concept                   | 0.119              | 0.120 | 0.112   | 0.112   | -0.002                      | -0.003  | -0.043* | -0.041*  | -1%                 | -3%  | -38% | -38% |
| discussion                | 0.121              | 0.116 | 0.105   | 0.098   | -0.010                      | -0.013* | -0.041* | -0.040*  | -7%                 | -17% | -35% | -35% |
| mean over templates       |                    |       |         |         | -0.012                      | -0.023  | -0.040  | -0.041   |                     |      |      |      |
| templates at $ z  \geq 2$ |                    |       |         |         | 5 of 10                     | 8 of 10 | 9 of 10 | 10 of 10 |                     |      |      |      |

(c) Extraction: the same 1000 prompts, only batch size and precision vary.

| batch size     | GPT-2 M             |                     | OLMo-1B             |                     |
|----------------|---------------------|---------------------|---------------------|---------------------|
|                | fp32 $\hat{\delta}$ | fp16 $\hat{\delta}$ | fp32 $\hat{\delta}$ | fp16 $\hat{\delta}$ |
| 1 (no padding) | 0.0816              | 0.0817              | 0.0817              | 0.0817              |
| 16             | 0.0996              | 0.0997              | 0.0817              | 0.0816              |
| 32             | 0.0905              | 0.0905              | 0.0817              | 0.0817              |

Table 12: **In text, clustered structure depends on recipe and scale: genuine at every Pythia scale and at the larger GPT-2 sizes, at the margin for GPT-2 S, absent for GPT-2 M and for the sentence embedders on this probe.** (a)  $r$  = replicates above the real value,  $p$  the left-tail add-one  $p$ -value; genuine = BH-corrected  $p \leq 0.05$  over the 15 models (°: not genuine); exc./null: excess as a fraction of the null reading; the  $2 \times 2$  column gives the verdicts under Haar $\times$ p99.9, Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum (● genuine); cosine: the same census on cosine geometry. Genuine 7/15; GPT-2 S is genuine at  $p = .020$  with excess  $-0.013$  and GPT-2 M is not ( $p = .154$ ,  $-0.005$ ), while L and XL are genuine under every construction. GTE-base has the highest effective rank and is not genuine, so genuineness is not a proxy for anisotropy; the only LLM-backed embedder (GTE-Qwen2-1.5B) is not genuine. (b) Causal-LM values are template-sensitive (range of raw  $\hat{\delta}$  up to 0.065 for causal LMs against  $\leq 0.012$  for embedders; “gloss only” contains no class name) but the scale ordering is template-robust: with only 3 null replicates per cell the starred S and M cells are borderline, as the 200-replicate record in (a) shows, whereas the L/XL cells are not in doubt. (c) Precision is irrelevant (fp16 equals fp32 to four decimals); for GPT-2, left-padded batching changes the hidden states (mean cosine to the padding-free extraction 0.98) and moves  $\hat{\delta}$  by up to 0.018, while OLMo handles left padding correctly and is invariant, which is why padding-free extraction is the record.

## B.12 IS LOCAL STRUCTURE SHARED? PERMUTATION-CALIBRATED AGREEMENT

| measure                          | raw   | null mean | $\tau_{0.05}$ | calibrated | null-centered | pairs $p < 0.05$ |
|----------------------------------|-------|-----------|---------------|------------|---------------|------------------|
| mutual-kNN ( $k=10$ ), Euclidean | 0.472 | 0.010     | 0.015         | 0.464      | 0.462         | 66/66            |
| mutual-kNN, Poincaré             | 0.425 | 0.010     | 0.019         | 0.414      | 0.415         | 66/66            |
| linear CKA, Euclidean            | 0.633 | 0.108     | 0.111         | 0.577      | 0.524         | 66/66            |
| linear CKA, Poincaré             | 0.640 | 0.108     | 0.111         | 0.586      | 0.532         | 66/66            |

Table 13: **Cross-model agreement survives permutation calibration.** Means over the 66 model pairs of the 12 vision backbones on the 1000 ImageNet class centroids (census cache). Calibration of Gröger et al. (2026) with  $K=200$  permutations of the class correspondence and  $\alpha=0.05$ , scalar (no layer search):  $\tau_{0.05}$  is the  $\lceil 0.95(K+1) \rceil$ -th order statistic of the observed score and its nulls (eq. 9),  $p = (1 + \#\{\text{null} \geq \text{obs}\}) / (K+1)$  (eq. 10), calibrated =  $\max\{(\text{obs} - \tau_{0.05}) / (1 - \tau_{0.05}), 0\}$  (eq. 12); “null-centered” is the earlier raw-minus-null-mean variant, kept for continuity. The mKNN null mean equals the analytic chance level  $k/(n-1) = 10/999$ .

## B.13 WHAT FOLLOWS FOR PRACTICE? THE COROLLARY

**One parameter-free projection and two tasks serve the corollary.** The parameter-free map  $\phi(x) = \tanh(\frac{1}{2}\|s(x - \mu)\|) \cdot \frac{s(x - \mu)}{\|s(x - \mu)\|}$  centers features, scales the 95th-percentile norm to radius  $1/\sqrt{2}$ , and preserves directions; prototypes in the ball are means of projected points re-mapped through the origin. Tasks: nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes); metric arms: Euclidean (R), Poincaré (H), the radial map with Euclidean distances (RT), cosine (COS), and the L2-normalized stack. Changing the readout metric on the intact configuration, which is what the rule selects among, is distinct from deforming the configuration itself.

**The calibrated reading predicts the gain within datasets.** Within each hierarchical dataset the Poincaré advantage tracks the record excess: the nearest-centroid correlation is  $-0.73$  on ImageNet with a confidence interval that excludes zero on all four datasets, the few-shot correlation excludes zero on CIFAR-100, CIFAR-10 and DTD but not on ImageNet, and the depth-test  $z$  predicts no gain anywhere (Table 14). The raw reading correlates as strongly ( $-0.84$  on ImageNet for nearest-centroid) and survives partialing out dimension, baseline accuracy, family and spectral covariates; without DINOv2 it holds on ImageNet and CIFAR-100 but not on CIFAR-10/DTD. Pooled cross-dataset correlations are family-driven and not used as evidence; on flat-label datasets no metric helps. Gains carry paired confidence intervals and a McNemar test.

**The training objective picks the metric.** The radial map with Euclidean distances changes nothing, so the gain is the metric itself, and the split follows the objective: cosine suffices for DINOv2, Poincaré  $\approx$  cosine for supervised ViTs, and Poincaré beats cosine for CLIP by up to  $+1.3\text{pp}$  even after L2 normalization (Figure 11; held-out selection and policy comparison in Table 14). The rule is read off the families it is scored on: it describes them rather than validating a policy.

**A diagnostic, not a policy.** Khrulkov et al. (2020) estimate  $c \approx 0.33$  from the raw reading of their datasets; the same rule on structureless Gaussian clouds gives the curvatures of Section 6. Cosine is a safe default, gating on the reading is the weakest policy tested, and gains are modest and limited to prototype-based tasks (sample-level tasks never benefit, Table 6). On DBpedia, a three-level text hierarchy, the diagnostic predicts correctly on both sides: genuine excess, no metric gain.

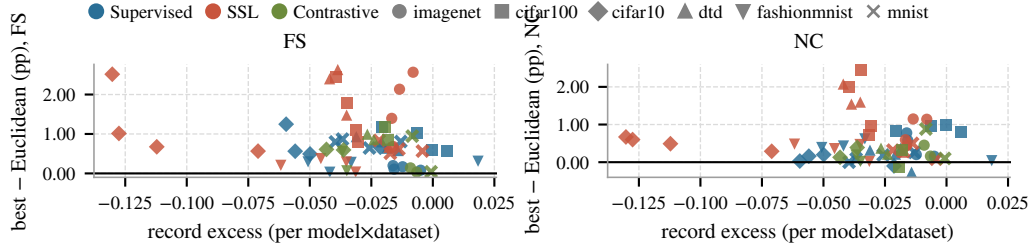


Figure 11: **Cells with a deeper calibrated excess carry larger zero-cost metric gains.** Plotted: best zero-cost metric minus Euclidean per cell against the record excess (FS left, NC right), one marker per dataset and family colors; the within-dataset correlations with confidence intervals are in Table 14.

(a) Task accuracies and zero-cost metric gains per cell (10 backbones with cached per-dataset features  $\times$  6 datasets).

| model    | data   | NC acc. (%) |        | FS acc. (%) |        | FS advantage over Euclidean (pp) |            |                           | McNemar        |
|----------|--------|-------------|--------|-------------|--------|----------------------------------|------------|---------------------------|----------------|
|          |        | Eucl.       | Poinc. | Eucl.       | Poinc. | cosine                           | radial map | Poinc. <sub>N</sub> - cos | NC H vs R      |
| ViT-T    | IN     | 68.1        | 67.2   | 97.5        | 97.6   | -0.01                            | +0.01      | +0.08                     | $10^{-22}$ (−) |
|          | C100   | 29.3        | 28.2   | 62.4        | 63.1   | +1.01                            | +0.45      | +0.55                     | $10^{-5}$ (−)  |
|          | C10    | 87.5        | 87.4   | 87.6        | 88.1   | +0.66                            | +0.19      | +0.29                     | 0.19 (−)       |
|          | DTD    | 63.7        | 62.5   | 83.3        | 83.9   | -0.23                            | +0.32      | +0.69                     | 0.03 (−)       |
|          | FMNIST | 78.5        | 78.4   | 81.6        | 81.9   | +0.21                            | +0.10      | +0.31                     | 0.38 (−)       |
|          | MNIST  | 75.9        | 74.0   | 74.3        | 74.6   | +0.64                            | +0.52      | +0.20                     | $10^{-13}$ (−) |
| ViT-S    | IN     | 77.7        | 77.5   | 98.6        | 98.7   | +0.17                            | -0.02      | -0.07                     | 0.09 (−)       |
|          | C100   | 74.9        | 74.6   | 92.2        | 92.4   | +0.59                            | +0.16      | -0.00                     | 0.10 (−)       |
|          | C10    | 93.4        | 93.2   | 92.0        | 92.5   | +0.56                            | +0.27      | +0.19                     | 0.02 (−)       |
|          | DTD    | 69.3        | 68.5   | 85.4        | 86.0   | -0.03                            | +0.25      | +0.73                     | 0.10 (−)       |
|          | FMNIST | 80.7        | 80.4   | 83.7        | 83.5   | +0.28                            | +0.08      | -0.19                     | 0.08 (−)       |
|          | MNIST  | 84.8        | 84.1   | 80.5        | 81.4   | +0.34                            | +0.65      | +0.84                     | $10^{-4}$ (−)  |
| ViT-B    | IN     | 81.6        | 81.7   | 98.9        | 99.0   | +0.20                            | -0.04      | -0.05                     | 0.21 (+)       |
|          | C100   | 77.3        | 77.0   | 92.5        | 92.7   | +0.56                            | +0.12      | -0.05                     | 0.07 (−)       |
|          | C10    | 94.5        | 94.0   | 92.7        | 93.0   | +0.49                            | +0.18      | +0.15                     | $10^{-5}$ (−)  |
|          | DTD    | 70.8        | 70.8   | 85.4        | 86.2   | +0.28                            | +0.19      | +0.69                     | 1.00 (−)       |
|          | FMNIST | 81.3        | 81.6   | 83.4        | 83.3   | +0.03                            | +0.09      | -0.04                     | 0.08 (+)       |
|          | MNIST  | 81.6        | 80.5   | 76.8        | 77.6   | +0.19                            | +0.51      | +0.66                     | $10^{-7}$ (−)  |
| ViT-L    | IN     | 86.1        | 86.2   | 99.4        | 99.4   | +0.11                            | -0.03      | -0.02                     | 0.58 (+)       |
|          | C100   | 85.3        | 85.2   | 94.8        | 95.5   | +1.17                            | +0.31      | +0.12                     | 0.59 (−)       |
|          | C10    | 97.5        | 97.5   | 95.0        | 95.8   | +1.25                            | +0.35      | +0.59                     | 0.76 (+)       |
|          | DTD    | 71.8        | 71.5   | 85.6        | 86.5   | +0.13                            | +0.16      | +1.02                     | 0.69 (−)       |
|          | FMNIST | 83.7        | 83.5   | 84.4        | 84.7   | +0.00                            | +0.17      | +0.23                     | 0.12 (−)       |
|          | MNIST  | 84.2        | 83.6   | 80.8        | 81.6   | +0.08                            | +0.42      | +0.67                     | 0.00 (−)       |
| DINOv2-S | IN     | 74.7        | 74.4   | 97.4        | 97.9   | +0.68                            | +0.03      | -0.22                     | 0.00 (−)       |
|          | C100   | 73.4        | 74.4   | 94.3        | 95.1   | +0.80                            | +0.15      | +0.06                     | $10^{-6}$ (+)  |
|          | C10    | 95.0        | 95.2   | 92.9        | 93.5   | -0.36                            | +0.13      | +1.06                     | 0.14 (+)       |
|          | DTD    | 73.4        | 73.4   | 89.8        | 90.5   | +0.30                            | +0.04      | +0.36                     | 1.00 (+)       |
|          | FMNIST | 82.2        | 82.2   | 85.4        | 85.5   | +0.02                            | +0.07      | +0.07                     | 1.00 (−)       |
|          | MNIST  | 83.4        | 81.9   | 80.2        | 80.8   | -0.07                            | +0.45      | +0.67                     | $10^{-11}$ (−) |
| DINOv2-B | IN     | 78.9        | 78.9   | 97.0        | 97.8   | +1.40                            | +0.04      | -0.63                     | 0.57 (+)       |
|          | C100   | 87.4        | 87.9   | 96.5        | 97.2   | +1.11                            | +0.09      | -0.34                     | $10^{-4}$ (+)  |
|          | C10    | 97.1        | 97.4   | 94.7        | 95.4   | +0.27                            | +0.03      | +0.43                     | $10^{-4}$ (+)  |
|          | DTD    | 74.2        | 74.7   | 89.2        | 90.3   | +1.48                            | +0.01      | -0.35                     | 0.31 (+)       |
|          | FMNIST | 83.2        | 82.8   | 85.9        | 85.9   | +0.21                            | +0.04      | -0.18                     | 0.01 (−)       |
|          | MNIST  | 80.6        | 78.8   | 76.6        | 77.1   | +0.17                            | +0.48      | +0.47                     | $10^{-16}$ (−) |
| DINOv2-L | IN     | 79.8        | 80.4   | 96.2        | 97.1   | +2.13                            | +0.02      | -1.24                     | $10^{-14}$ (+) |
|          | C100   | 85.6        | 87.7   | 96.1        | 97.1   | +1.80                            | +0.09      | -0.74                     | $10^{-38}$ (+) |
|          | C10    | 97.9        | 98.3   | 94.1        | 95.1   | +0.92                            | -0.00      | +0.11                     | $10^{-8}$ (+)  |
|          | DTD    | 74.4        | 75.2   | 87.1        | 88.4   | +2.40                            | +0.03      | -0.98                     | 0.11 (+)       |
|          | FMNIST | 84.4        | 84.5   | 87.8        | 87.9   | +0.29                            | +0.03      | -0.03                     | 0.40 (+)       |
|          | MNIST  | 74.1        | 72.7   | 70.5        | 71.1   | +0.25                            | +0.43      | +0.46                     | $10^{-12}$ (−) |
| DINOv2-G | IN     | 78.4        | 78.3   | 96.2        | 97.1   | +2.57                            | +0.03      | -1.65                     | 0.10 (−)       |
|          | C100   | 89.8        | 91.0   | 95.9        | 97.2   | +2.44                            | +0.13      | -1.10                     | $10^{-16}$ (+) |
|          | C10    | 97.8        | 98.2   | 92.3        | 93.8   | +2.51                            | -0.02      | -1.01                     | $10^{-13}$ (+) |
|          | DTD    | 72.9        | 73.1   | 87.2        | 88.5   | +2.63                            | +0.01      | -1.38                     | 0.81 (+)       |
|          | FMNIST | 85.9        | 85.8   | 89.0        | 89.2   | +0.38                            | +0.04      | -0.07                     | 0.72 (−)       |
|          | MNIST  | 79.3        | 78.6   | 72.9        | 73.8   | +0.30                            | +0.56      | +0.72                     | 0.00 (−)       |
| CLIP-B   | IN     | 61.4        | 59.9   | 96.8        | 96.8   | -0.04                            | -0.00      | +0.14                     | $10^{-40}$ (−) |
|          | C100   | 65.6        | 65.1   | 87.7        | 88.9   | -0.03                            | +0.44      | +1.21                     | 0.07 (−)       |
|          | C10    | 87.9        | 88.3   | 89.5        | 90.1   | -0.19                            | +0.20      | +0.90                     | $10^{-4}$ (+)  |
|          | DTD    | 63.7        | 63.2   | 82.2        | 83.1   | -0.23                            | +0.36      | +1.29                     | 0.46 (−)       |
|          | FMNIST | 75.9        | 75.1   | 78.8        | 78.9   | +0.06                            | +0.17      | +0.10                     | $10^{-4}$ (−)  |
|          | MNIST  | 70.1        | 66.6   | 70.9        | 69.7   | +0.04                            | +0.23      | -1.17                     | $10^{-45}$ (−) |
| CLIP-L   | IN     | 75.0        | 74.5   | 98.4        | 98.6   | +0.01                            | +0.00      | +0.22                     | $10^{-8}$ (−)  |
|          | C100   | 74.6        | 74.5   | 90.0        | 90.8   | +0.12                            | +0.39      | +1.23                     | 0.66 (−)       |
|          | C10    | 95.9        | 95.9   | 92.4        | 93.0   | +0.16                            | +0.25      | +0.91                     | 0.59 (−)       |
|          | DTD    | 68.6        | 69.0   | 85.2        | 86.2   | -0.08                            | +0.36      | +1.24                     | 0.45 (+)       |
|          | FMNIST | 76.2        | 75.9   | 76.8        | 77.3   | +0.12                            | +0.33      | +0.68                     | 0.09 (−)       |
|          | MNIST  | 85.2        | 86.1   | 83.7        | 84.7   | +0.08                            | +0.58      | +1.06                     | $10^{-7}$ (+)  |

Table 14: **The calibrated reading predicts the zero-cost gain within datasets, the depth verdict predicts nothing, and cosine is the safe default.** (a) Nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes) accuracy under Euclidean and Poincaré readouts, regenerated from the audited reruns; FS advantage of cosine and of the radial map with Euclidean distances (the map without the metric changes nothing) over Euclidean, and of the Poincaré readout over cosine after L2 normalization; McNemar  $p$  for the NC test-set decisions, Poincaré vs Euclidean (order of magnitude when  $p < 10^{-3}$ ; sign of the NC advantage in parentheses).

(b) Correlation between the raw supremum  $\delta_{\text{norm}}$  and the metric gain within each dataset ( $n=10$  backbones), with bootstrap CI95 and the family-identity control (demeaned: both variables centered within family; within-family  $r$  for ViT and DINOv2, sign only for CLIP).

| gain      | dataset                    | $r$ [CI95]           | demeaned | ViT   | DINOv2 | CLIP |
|-----------|----------------------------|----------------------|----------|-------|--------|------|
| NC H—R    | ImageNet                   | −0.83 [−0.97, −0.65] | −0.73    | −0.99 | −0.77  | (−1) |
|           | CIFAR-100                  | −0.87 [−0.97, −0.78] | −0.45    | −0.68 | −0.33  | (−1) |
|           | CIFAR-10                   | −0.63 [−0.97, −0.06] | −0.31    | +0.01 | −0.98  | (+1) |
|           | DTD                        | −0.46 [−0.90, +0.22] | −0.57    | −0.78 | −0.76  | (+1) |
|           | pooled (cluster bootstrap) | −0.45 [−0.60, −0.36] | —        | —     | —      | —    |
| FS H—R    | ImageNet                   | −0.57 [−0.91, +0.14] | −0.73    | −0.04 | −0.97  | (−1) |
|           | CIFAR-100                  | −0.67 [−0.95, −0.15] | −0.63    | −0.41 | −0.96  | (+1) |
|           | CIFAR-10                   | −0.78 [−0.93, −0.23] | −0.64    | −0.22 | −0.79  | (−1) |
|           | DTD                        | −0.71 [−0.94, −0.32] | −0.86    | −0.95 | −0.98  | (+1) |
|           | pooled (cluster bootstrap) | −0.18 [−0.50, +0.12] | —        | —     | —      | —    |
| FS best—R | ImageNet                   | −0.64 —              | −0.71    | −0.57 | −0.90  | (−1) |
|           | CIFAR-100                  | −0.83 —              | −0.82    | −0.58 | −0.96  | (+1) |
|           | CIFAR-10                   | −0.63 —              | −0.55    | −0.26 | −0.65  | (−1) |
|           | DTD                        | −0.60 —              | −0.70    | −0.95 | −0.88  | (+1) |

(c) Best zero-cost metric per backbone: few-shot advantage over Euclidean, mean over the four hierarchical datasets, and which metric collects it (either: within 0.15pp).

| model    | best—R (pp) | metric |
|----------|-------------|--------|
| ViT-T    | +0.58       | either |
| ViT-S    | +0.49       | either |
| ViT-B    | +0.51       | either |
| ViT-L    | +0.87       | either |
| DINOv2-S | +0.68       | H      |
| DINOv2-B | +1.17       | cos    |
| DINOv2-L | +1.84       | cos    |
| DINOv2-G | +2.54       | cos    |
| CLIP-B   | +0.67       | H      |
| CLIP-L   | +0.65       | H      |

Table 14: (continued) (b) Within-dataset correlations of the raw supremum reading; bootstrap CI95 from 10k resamples of the 10 models, pooled rows by cluster bootstrap over datasets; family control: both variables centered within family before correlating, and within-family correlations (ViT, DINOv2: 4 models each; CLIP has two, so only its sign is shown). Without DINOv2 (six backbones) the raw NC correlation holds on ImageNet and CIFAR-100 (−0.87/−0.65) but not on CIFAR-10/DTD (+0.23/−0.34); pooled cross-dataset correlations of the best-metric advantage do not survive the family control (−0.31 → −0.18 over all 60 cells, −0.33 → −0.04 on hierarchical cells) and are family-driven. On the record excess the nearest-centroid prediction survives (CI95 excluding zero) on ImageNet, CIFAR-100, CIFAR-10, DTD and the few-shot prediction on CIFAR-100, CIFAR-10, DTD; the depth verdict predicts no gain in any dataset. (c) The gain per backbone and which readout collects it.

(d) The same correlations on the raw  $\hat{\delta}_{99.9}$ , the record excess and the depth-test  $z$  (Fisher- $z$  CI95; dm: family-demeaned  $r$ ).

| gain      | dataset   | raw $\hat{\delta}_{99.9}$ |       | record excess        |       | depth $z$            |  |
|-----------|-----------|---------------------------|-------|----------------------|-------|----------------------|--|
|           |           | $r$ [CI95]                | dm    | $r$ [CI95]           | dm    | $r$ [CI95]           |  |
| NC H—R    | ImageNet  | −0.84 [−0.96, −0.45]      | −0.83 | −0.73 [−0.93, −0.19] | −0.61 | −0.28 [−0.77, +0.42] |  |
|           | CIFAR-100 | −0.92 [−0.98, −0.68]      | −0.58 | −0.79 [−0.95, −0.33] | −0.21 | +0.52 [−0.16, +0.87] |  |
|           | CIFAR-10  | −0.63 [−0.90, +0.00]      | −0.31 | −0.67 [−0.91, −0.06] | −0.34 | n/a                  |  |
|           | DTD       | −0.67 [−0.91, −0.07]      | −0.67 | −0.76 [−0.94, −0.26] | −0.60 | n/a                  |  |
| FS H—R    | ImageNet  | −0.86 [−0.97, −0.50]      | −0.56 | −0.21 [−0.74, +0.48] | +0.35 | +0.27 [−0.43, +0.77] |  |
|           | CIFAR-100 | −0.62 [−0.90, +0.02]      | −0.51 | −0.79 [−0.95, −0.31] | −0.80 | −0.26 [−0.76, +0.45] |  |
|           | CIFAR-10  | −0.78 [−0.95, −0.31]      | −0.64 | −0.78 [−0.95, −0.30] | −0.66 | n/a                  |  |
|           | DTD       | −0.87 [−0.97, −0.53]      | −0.94 | −0.98 [−1.00, −0.91] | −0.97 | n/a                  |  |
| FS best—R | ImageNet  | −0.83 [−0.96, −0.43]      | −0.50 | −0.08 [−0.68, +0.58] | +0.47 | +0.13 [−0.54, +0.70] |  |
|           | CIFAR-100 | −0.77 [−0.94, −0.27]      | −0.75 | −0.75 [−0.94, −0.23] | −0.65 | +0.22 [−0.48, +0.75] |  |
|           | CIFAR-10  | −0.63 [−0.90, −0.01]      | −0.55 | −0.62 [−0.90, +0.02] | −0.56 | n/a                  |  |
|           | DTD       | −0.81 [−0.95, −0.36]      | −0.78 | −0.90 [−0.98, −0.63] | −0.84 | n/a                  |  |

(e) Few-shot metric-selection policies on held-out episodes (500 validation / 500 test per cell).

| policy (test advantage over Euclidean, pp) | all 60 cells | hierarchical (40) |
|--------------------------------------------|--------------|-------------------|
| always cosine                              | +0.23        | +0.28             |
| objective rule (zero validation)           | +0.28        | +0.41             |
| $\delta$ -gated objective rule             | +0.08        | +0.13             |
| validation-picked per cell                 | +0.63        | +0.78             |
| test oracle (reference)                    | +0.64        | +0.78             |

(f) Nearest-centroid metric-selection policies.

| NC policy (advantage over Euclidean, pp) | all 60 | flat (20) | hierarchical (40) |
|------------------------------------------|--------|-----------|-------------------|
| always cosine                            | +0.48  | +0.24     | +0.59             |
| always Poincaré                          | −0.24  | −0.70     | −0.00             |
| gated rule (flat→R)                      | +0.34  | +0.00     | +0.51             |
| $\delta$ -gated rule                     | +0.19  | +0.00     | +0.29             |

(g) Sensitivity to the projection target  $t$  (36 hierarchical cells).

| target $t$        | 0.25  | 0.35  | 0.41  | 0.45  | 0.50  | 0.58  | 0.71  | 0.95  |
|-------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| FS adv. mean (pp) | +0.88 | +0.87 | +0.87 | +0.86 | +0.84 | +0.82 | +0.76 | +0.56 |
| NC adv. mean (pp) | −0.30 | −0.25 | −0.23 | −0.21 | −0.17 | −0.12 | −0.01 | +0.15 |
| % cells FS > 0    | 97    | 100   | 100   | 100   | 100   | 100   | 100   | 100   |

Table 14: (continued) (d) The calibrated reading: the raw 99.9th-percentile statistic, the record excess and the depth-test  $z$  of Table 8 as predictors. (e) Validation picking retains the oracle advantage and the zero-validation objective rule modestly beats always-cosine. (f) The gate’s value is avoiding the Poincaré projection on flat-label data; cosine is a safe NC default everywhere. (g) The headline  $t = 1/\sqrt{2} \approx 0.71$  was fixed a priori and is conservative for FS.

#### B.14 DOES $\xi$ READ CURVATURE? THE PARALLELOGRAM DEFECT AS A NORM-STRUCTURE DESCRIPTOR

**$\xi$  reads norm structure, not curvature.** A parallelogram estimator  $\xi$ , intended as a curvature sign, is driven negative by heterogeneous norms alone, so it is reported only as a norm-structure descriptor. For a triangle  $(a, b, c)$  with  $m$  the approximate midpoint of  $(b, c)$  (the minimizer of  $D(\cdot, b) + D(\cdot, c) + |D(\cdot, b) - D(\cdot, c)|$ ), the parallelogram defect is  $\xi = (D(a, m)^2 + \frac{1}{4}D(b, c)^2 - \frac{1}{2}D(a, b)^2 - \frac{1}{2}D(a, c)^2) / (2 D(a, m) \text{diam})$ , averaged over  $2 \times 10^4$  sampled triangles: 0 in Euclidean space, positive on spheres, negative in hyperbolic space and on trees.

(a)  $\xi$  on reference geometries and ImageNet centroids, with the spectrum-matched null (20 replicates).

| space / model                   | $\xi$ (mean) | frac. negative | null $\xi$         | excess ( $\pm$ CI95) | $z$ |
|---------------------------------|--------------|----------------|--------------------|----------------------|-----|
| iid Gaussian ( $d=768$ )        | +0.092       | 0%             | —                  | —                    | —   |
| $S^d$ (geodesic)                | +0.113       | 0%             | —                  | —                    | —   |
| $S^d$ (chord)                   | +0.115       | 0%             | —                  | —                    | —   |
| $\mathbb{H}^2$ region ( $R=6$ ) | -0.194       | 96%            | —                  | —                    | —   |
| balanced tree (depth 9)         | -0.239       | 96%            | —                  | —                    | —   |
| ViT-T                           | +0.027       | 25%            | +0.027 $\pm$ 0.002 | -0.001 $\pm$ 0.003   | -0  |
| ViT-S                           | +0.027       | 23%            | +0.042 $\pm$ 0.002 | -0.015 $\pm$ 0.005   | -7  |
| ViT-B                           | +0.033       | 16%            | +0.047 $\pm$ 0.002 | -0.014 $\pm$ 0.004   | -6  |
| ViT-L                           | +0.028       | 21%            | +0.046 $\pm$ 0.002 | -0.018 $\pm$ 0.005   | -7  |
| DINO-B                          | -0.008       | 55%            | +0.052 $\pm$ 0.003 | -0.059 $\pm$ 0.005   | -21 |
| DINOv2-S                        | -0.020       | 66%            | +0.061 $\pm$ 0.004 | -0.081 $\pm$ 0.008   | -20 |
| DINOv2-B                        | -0.049       | 83%            | +0.075 $\pm$ 0.003 | -0.123 $\pm$ 0.005   | -45 |
| DINOv2-L                        | -0.068       | 90%            | +0.080 $\pm$ 0.002 | -0.149 $\pm$ 0.004   | -66 |
| DINOv2-G                        | -0.070       | 89%            | +0.082 $\pm$ 0.002 | -0.152 $\pm$ 0.004   | -69 |
| CLIP-B                          | +0.019       | 32%            | +0.024 $\pm$ 0.002 | -0.005 $\pm$ 0.003   | -3  |
| CLIP-L                          | +0.028       | 28%            | +0.032 $\pm$ 0.002 | -0.004 $\pm$ 0.004   | -2  |
| SigLIP-B                        | +0.036       | 20%            | +0.031 $\pm$ 0.002 | +0.005 $\pm$ 0.004   | +2  |

(b)  $\xi$  on the angular geometry: centroids L2-normalized, spherical geodesic distances, null on the normalized cloud.

| model    | $\xi_{\text{geo}}$ | frac. neg. | excess  | $z$   |
|----------|--------------------|------------|---------|-------|
| ViT-T    | +0.031             | 0.24       | -0.0048 | -2.7  |
| ViT-S    | +0.065             | 0.08       | +0.0092 | +6.2  |
| ViT-B    | +0.067             | 0.07       | +0.0144 | +10.5 |
| ViT-L    | +0.074             | 0.04       | +0.0113 | +7.9  |
| DINO-B   | +0.085             | 0.05       | +0.0014 | +1.0  |
| DINOv2-S | +0.100             | 0.02       | +0.0033 | +2.3  |
| DINOv2-B | +0.108             | 0.01       | +0.0046 | +4.9  |
| DINOv2-L | +0.108             | 0.01       | +0.0040 | +3.9  |
| DINOv2-G | +0.094             | 0.01       | -0.0114 | -12.5 |
| CLIP-B   | +0.025             | 0.28       | -0.0129 | -7.1  |
| CLIP-L   | +0.033             | 0.25       | -0.0120 | -4.8  |
| SigLIP-B | +0.051             | 0.14       | +0.0025 | +1.2  |

(c)  $\xi$  on references with heterogeneous norms but no curvature ( $n=1000$ ,  $d=768$ , 5 seeds).

| reference (no curvature unless stated)     | $\xi$   | frac. negative |
|--------------------------------------------|---------|----------------|
| iid Gaussian                               | +0.0919 | 0.00           |
| Gaussian, lognormal radii ( $\sigma=0.3$ ) | -0.0621 | 0.99           |
| Gaussian, lognormal radii ( $\sigma=0.6$ ) | -0.0607 | 1.00           |
| Gaussian, lognormal radii ( $\sigma=1.0$ ) | -0.0496 | 0.99           |
| core + shell mixture                       | -0.3217 | 0.79           |
| uniform sphere                             | +0.1154 | 0.00           |
| sphere with radial jitter                  | -0.0571 | 1.00           |

Table 15:  $\xi$  reads norm structure, not curvature. (a) Matched nulls have positive  $\xi$  (noise combines null and estimator s.d.), so raw readings understate curvature. (b) The deep negative  $\xi$ -excess of DINOv2 on ImageNet is a norm-structure effect for S/B/L, whose angular  $\xi$ -excess turns sphere-leaning positive, while DINOv2-G and CLIP-B/L remain negative on the sphere. (c) Norm dispersion alone drives  $\xi$  negative (lognormal radii, radial jitter on a sphere, core+shell), so the estimator is reported only as a norm-structure descriptor.

## B.15 PROVENANCE

| table | generated file        | result file(s)                                                                                                                                                                                                                                                                                                                                   |
|-------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4     | tab_q01_census        | expR52_census_haar_p999_200.csv<br>expR54_census_haar_sup_200.csv expR40b_p999census200.csv<br>expR39c_census200_cache.csv<br>expR57_census_cosine_haar_p999_200.csv<br>expR57_text_cosine_haar_p999_200.csv<br>exp10_local_vs_global.csv expR45_convnet_rows.csv<br>exp20_null_ztable.csv                                                       |
| 12    | tab_q02_text          | expR53_text_haar_p999_200.csv expR53_text_haar_sup_200.csv<br>expR53_text_gauss_p999_200.csv expR48b_text_census200_bs1.csv<br>expR57_text_cosine_haar_p999_200.csv exp18_text_nulls.csv<br>exp4_prompt_variation.csv exp17_wordnet_prompts.csv<br>expR49_template_nulls_bs1.csv expR47_extraction_variance.csv<br>expR62_samplelevel_record.csv |
| 6     | tab_q03_sample        | expR56_depth_variants.csv expR69_depth_haarhubs_summary.csv                                                                                                                                                                                                                                                                                      |
| 8     | tab_q04_depth         | expR64b_wn30bal_summary.csv expR70_inet1k_supervised.csv<br>expR52_census_haar_p999_200.csv expR63_meru_record.csv<br>expR65_hier_finetune.csv expR71_meru_radii.csv                                                                                                                                                                             |
| 9     | tab_q05_power         | expR55_depth_power.csv expR55b_depth_power_leafframe.csv<br>expR64b_wn30.csv expR64b_wn30_summary.csv<br>expR64b_wn30bal_summary.csv expR67_frame_choice.json                                                                                                                                                                                    |
| 10    | tab_q06_treemap       | expR69_depth_haarhubs_summary.csv<br>exp22_tree_similarity_imagenet.npz exp23_treemap_controls.npz<br>exp23_config_diagnostics.json                                                                                                                                                                                                              |
| 11    | tab_q07_wordnet       | expR58_treemap_cutfree_summary.csv<br>exp3_alignment.csv exp28_recovery_per_config.csv<br>exp8_p1_recovery.csv exp1_delta_controls.csv<br>exp8_p6_pooling.csv expR70_inet1k_supervised.csv<br>exp14_dbpedia.csv expR61_dbpedia_record.csv<br>exp27_dbpedia_treemap.json exp16_hierarcaps.csv                                                     |
| 2     | tab_q08_robust        | expR72_budget_record.csv expR72_budget_record_summary.csv<br>expR31_quad_sweep.csv expR32_centroid_bootstrap.csv<br>expR46_null_variants.csv<br>expR59_imagenet_bootstrap_summary.csv<br>expR66_joint_sensitivity_summary.csv<br>expR60_c_sweep_record.csv                                                                                       |
| 14    | tab_q09_corollary     | table1_regenerated.csv exp2_metric_controls.csv<br>exp2b_normalized_stack.csv exp13_mcnemar.csv<br>night/correlation_cis.csv exp20_null_ztable.csv<br>expR68_corollary_excess.csv exp24_val_metric_selection.csv<br>night/t_sweep.csv                                                                                                            |
| 3     | tab_q10_calibration   | exp1_delta_controls.csv expR42_star_calibration.csv                                                                                                                                                                                                                                                                                              |
| 7     | tab_q11_interventions | analysis4_finetuning.csv e1_delta_by_layer.csv<br>e5_random_control.csv night/arch_matched_triplet.csv<br>night/orc_bridges.csv                                                                                                                                                                                                                  |
| 15    | tab_q12_xi            | exp12_curvature_sign.csv exp26_xi_nulls.csv<br>expR38_xi_angular.csv expR43_xi_norm_references.csv                                                                                                                                                                                                                                               |
| 13    | tab_q13_local         | exp21b_local_global_K200.csv                                                                                                                                                                                                                                                                                                                     |
| 5     | tab_q14_panel         | (model panel: stated in the generator; parameter counts<br>from the model cards)                                                                                                                                                                                                                                                                 |

Table 16: Provenance index: every appendix table is generated by script from the released per-cell result files listed here; no table value is hand-transcribed. The same mapping for main-text tables and figures is annotated in the source and shipped with the code release.